

Expander-Based Codes at the Gilbert–Varshamov Bound

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Abstract

We prove that Expand–Accumulate (EA) codes approach the Gilbert–Varshamov (GV) distance as the code length grows. The result holds over the binary field and every fixed finite field. At fixed rate, encoding uses $O(n \log n)$ expected field operations, with a tunable inverse-polynomial probability of sampling a bad code. Fixed-memory wrapped binary Expand–Convolute (EC) codes satisfy the same guarantee. Allowing $O(n \log^2 n)$ expected field operations makes sampling failure negligible in the code length.

These codes combine a sparse random linear map with an invertible recursive map. Our analysis uses exact input–output weight enumerators in place of the earlier concentration bounds. It also gives tighter constants in the logarithmic expected degree. For EA-based pseudorandom correlation functions, these bounds improve the degree–distance tradeoff that controls local evaluation cost.

Separately, we obtain finite-length bounds at small degrees by replacing independent incidence with regional regularity. At rate one half, a two-sided regular EC ensemble with left/right degrees 10/5 and memory 15 reaches 99.993% of the binary Gilbert–Varshamov distance. The probability of sampling a generator below this distance is less than $2^{-32.4990}$.

We extend the construction to finite fields using independent nonzero edge labels and full-field convolution coefficients. These coefficients mix field components; scalar extension of a binary matrix does not. Over $\mathbb{F}_{2^{128}}$, degrees 24/12 with memory 5 reach 98.415% of the rate-one-half Gilbert–Varshamov distance with sampling failure below $2^{-45.03}$. Outward-rounded verifiers certify every finite claim. The results concern minimum distance of sampled ensembles; they do not provide a decoder or a complete protocol-security proof.¹

1 Introduction

Sparse linear codes appear inside pseudorandom correlation generators (PCGs), pseudorandom correlation functions (PCFs), and silent cryptographic protocols. A PCF lets two parties use short correlated keys to generate correlated outputs locally, on demand, at a shared input. These applications do not begin with a received word and ask a decoder to correct errors. They instead need a sampled linear map to send every nonzero message to an output with many nonzero coordinates. In coding terminology, the generator must have large minimum distance.

Expand–Accumulate (EA) and Expand–Convolute (EC) are two such families introduced for cryptographic applications [BCG⁺22, RRT23]. Both first apply a sparse random matrix. EA follows that matrix by a running sum. EC follows it by a randomized recursive convolution. The sparse first stage keeps encoding cheap, while the recursive second stage spreads each nonzero contribution through the output.

Impact on EA-based correlation functions. An important application is the EA-based PCF of Boyle et al. [BCG⁺22]. Its local evaluation uses one distributed comparison-function evaluation

¹This paper was created with the assistance of ChatGPT 5.6, 6.0.

per nonzero expander entry in the requested row. The comparison primitive gives the parties shares of a value determined by a hidden threshold. The single accumulator is essential to this construction: accumulated sparse noise can be represented by threshold functions. Our EA bounds improve the provable degree–distance tradeoff without replacing that accumulator. They therefore target a cost paid at every PCF evaluation, not just the cost of expanding a complete batch. Section 9 gives the cost connection and distinguishes the improvement in proved bounds from the original empirical parameter choices.

One accumulation stage. EA also has a simple parallel encoding structure: sparse expansion followed by one parallel prefix sum. Our improved distance guarantees retain this structure. Repeat–accumulate–accumulate (RAA) codes instead compose two accumulation stages, with a permutation between them [BCG⁺22, Remark 3.4]. EA therefore avoids a second dependent scan of the full vector. Both constructions admit logarithmic arithmetic depth; the advantage is fewer sequential stages, not a different asymptotic depth class. Section 2.3 states the cost model, including transposed encoding.

Full asymptotic GV distance at logarithmic degree. Our main asymptotic result reaches the Gilbert–Varshamov (GV) distance, the standard volume-based benchmark for random linear codes. For every desired failure exponent $A > 0$, a sufficiently large constant C gives sampling failure at most n^{-A} for all sufficiently large n . For EA over every fixed finite field, expected expander degree $C \ln n$ gives a vanishing gap of at most $1/\ln n + 1/n$ below GV. The same guarantee holds for binary wrapped EC with any fixed memory. At fixed rate, encoding uses $O(n \log n)$ expected field operations. Theorem 7.5 proves this claim, with C depending on the field, limiting rate, desired failure exponent, and recursive map.

For negligible sampling failure, increase the expected degree to $(\ln n)^2$. Corollary 7.6 retains the same distance target and gives failure $\exp(-\Omega((\ln n)^2))$, smaller than every inverse polynomial in n . Its expected encoding cost is $O(n \log^2 n)$ field operations. The extra logarithmic factor strengthens the failure guarantee; it is not needed to approach GV.

The full-field EC construction is instead covered by the finite certificates below. Those finite-length certificates are distinct from this growing-degree asymptotic analysis. The finite-length tables report explicit expander degrees and convolution memories to expose the concrete costs.

The published distance proofs do not analyze the recursive output law exactly. They compare or bound a state process and then apply concentration. This choice obscures two different regimes. A message with linearly many nonzero symbols makes the expander output nearly uniform; its distance is governed by the same entropy calculation as a random linear code. A message with only a few nonzero symbols instead tests whether the sparse first stage creates enough opportunities for the recursive map to become active. The earlier bounds lose substantially in both regimes.

This paper keeps the exact input–output weight enumerator of the recursive map. An enumerator is a generating function that records how many inputs of each weight produce outputs of each weight. Transfer matrices compute this function while retaining only the information about the current state that can affect future output weight. The accumulator needs two states. A convolution of memory m needs only $m + 1$ states, rather than all p^m possible field-valued states.

Two cases. We first study binary codes. This case models bit-valued choices, as in Silent OT, and the Hamming support determines every message value. We then study codes over a finite field $\mathbb{F}_p$ whose order is close to 2^{128} . This case models field-valued correlations, as in Silent VOLE. Here

equal supports can cancel differently, so the construction and the proof must mix field values. The two cases share the enumerator method but have different sparse-message bottlenecks.

Binary analysis of the original ensembles. Our first results change only the analysis. For the binary Bernoulli EA ensemble, write the entry probability as $C \ln(n)/n$, so an expander row has expected weight $C \ln n$. The exact transfer bound lowers the required constant at relative distance 0.05—meaning output weight $0.05n$ —from $C > 4.938$ to $C > 1.773$. At rate $1/5$, the dense-message condition improves from a distance below 0.1127 to the binary GV value 0.2430. For the wrapped EC construction already proposed in [RRT23], our theorem permits fixed memory 21, $C > 1.111623$, and $R < 0.7136$ at distance 0.05. The published nonwrapping theorem instead uses memory proportional to $\log n$ and imposes $C > 10.4344$ and $R < 0.1278$ among its conditions. Thus this asymptotic comparison also changes the analyzed EC variant. Our finite comparison in Section 5 holds the wrapped ensemble fixed.

The GV value is a benchmark defined by the volume of a Hamming ball. It is not a finite-length upper bound. Saying that a certificate reaches the floored GV cutoff means that its integer distance threshold is the floor of the corresponding asymptotic value at that length.

Binary regular-expander results. The exact analysis identifies low-support messages as the remaining cost. We therefore replace independent edge incidence by a region-stratified left-regular expander. Each input coordinate has exactly one edge in each output region. This change is part of the construction, not merely a proof improvement. At rate one half, the regular binary EA certificate uses degree 64 instead of Bernoulli expected degree 75. The regular binary EC certificate uses degree 28 instead of Bernoulli expected degree 42.2127, with convolution memory 9. Fixing both expander degrees improves EC much further. A binary two-sided regular EC ensemble uses left/right degrees 10/5 and convolution memory 15. Its cutoff reaches 99.993% of the rate-one-half GV distance, with sampling failure below $2^{-32.4990}$. Two intermediate profiles use degrees 14/7 with memory 6 and degrees 18/9 with memory 4. Their sampling failures are below $2^{-26.0875}$ and $2^{-21.6746}$, respectively. A lower-degree profile uses degrees 6/3 and memory 79 at the same cutoff. Its sampling failure is below $2^{-21.4222}$.

Scalar extension does preserve the minimum distance of a binary generator, but it does not add field mixing. In a basis of $\mathbb{F}_{2^{128}}$ over $\mathbb{F}_2$, the resulting map is 128 independent copies of the binary map. This is the right structure for binary-choice Silent OT, but not the field-valued mixing needed by Silent VOLE over $\mathbb{F}_{2^{128}}$.

Fields near 128 bits. Let p be a prime power. Over the finite field $\mathbb{F}_p$, cancellations depend on message values as well as their support. We attach an independent nonzero field label to each expander edge. The label symmetry makes the distribution of a fixed message depend only on its support. Because nonzero scalar multiples have the same zero coordinates after every linear map, we count each equivalence class of scalar-multiple messages once. These classes are the one-dimensional message subspaces. Constraint traces then group the resulting projective union before it becomes too large.

For finite-field EC, we also sample each feedback vector from the full field. This distribution preserves an exact reduced state based only on the number of trailing zero outputs. At rate one half over $\mathbb{F}_{2^{127}-1}$, we certify the floored p -ary GV cutoff at length 2,097,150. The expander has left degree 22 and right degree 11, and the convolution has memory 12. The probability of sampling a generator below the claimed distance is less than $2^{-27.2951}$.

Table 1: Representative finite certificates. The probability is over the random choices used to sample the generator. The final column reports the distance, failure bound, or comparison relevant to each row.

ensemble	field	rate	expander	recursive map	conclusion
<i>Binary case</i>					
EA	$\mathbb{F}_2$	1/5	Bernoulli, mean 50	accumulator	$U < 2^{-20.46}$ at $\delta = 0.05$
wrapped EC	$\mathbb{F}_2$	1/5	Bernoulli, mean 46.42	memory 5	$U < 2^{-32.08}$; 10.01-bit gain
regular EA	$\mathbb{F}_2$	1/2	left degree 64	accumulator	99.988% of GV; 14.67% fewer edges
regular EC	$\mathbb{F}_2$	1/2	left degree 28	memory 9	99.994% of GV; 33.67% fewer edges
two-sided EC	$\mathbb{F}_2$	1/2	degrees 18/9	memory 4	99.993% of GV; $U < 2^{-21.674}$
two-sided EC	$\mathbb{F}_2$	1/2	degrees 14/7	memory 6	99.993% of GV; $U < 2^{-26.087}$
two-sided EC	$\mathbb{F}_2$	1/2	degrees 10/5	memory 15	99.993% of GV; $U < 2^{-32.499}$
two-sided EC	$\mathbb{F}_2$	1/2	degrees 6/3	memory 79	99.993% of GV; $U < 2^{-21.422}$
<i>Fields near 128 bits</i>					
two-sided EC	$\mathbb{F}_{2^{127}-1}$	1/2	labeled degrees 22/11	memory 12	floored p -ary GV; $U < 2^{-27.295}$
two-sided EC	$\mathbb{F}_{2^{128}}$	1/2	labeled degrees 24/12	memory 5	98.415% of GV; $U < 2^{-45.03}$
two-sided EC	$\mathbb{F}_{2^{128}}$	1/2	labeled degrees 26/13	memory 4	98.415% of GV; $U < 2^{-135.20}$

For the field-valued Silent VOLE setting, we separately instantiate the same construction over $\mathbb{F}_{2^{128}}$. Degrees 24/12 with memory 5 reach 98.415% of the rate-one-half GV distance. The certified sampling-failure probability is below $2^{-45.03}$. Degrees 26/13 with memory 4 reach the same distance with sampling failure below $2^{-135.20}$. Both proved ensembles use nonzero field labels on expander edges and full-field coefficients in the convolution.

Keeping degrees 24/12 and memory 5 fixed gives a local field-size profile. At the corresponding field-relative cutoffs, the certified failure exponents are 45.84, 45.03, 44.24, and 38.78 for field orders 2^{127} , 2^{128} , 2^{129} , and 2^{136} . The decline comes from the increasing distance target rather than weaker field mixing at a fixed target.

Contributions. The paper makes five contributions.

1. We give a common exact-enumerator framework for sparse random maps followed by recursive rate-one maps.
2. We sharpen the distance proofs for the original binary EA and wrapped EC ensembles, including direct comparisons with the published bounds. For EA, the improved degree bounds address the local evaluation cost of the existing PCF construction.
3. We analyze left-regular and two-sided regular expanders. Binary EC gives four near-GV degree-memory points.
4. We extend the construction and analysis to fields near 128 bits using edge labels, projective counting, and constraint traces. A certified local profile separates field-size effects from changes in the distance target.
5. We give reproducible finite certificates checked with outward-rounded ball arithmetic. The certificates cover every nonzero message support.

Scope. Our probability bounds concern the sampling of a generator. They do not analyze a decoder, a noise distribution, or a complete cryptographic protocol. Each large-field headline result fixes one field and one finite length. Increasing the field order also changes the field-relative GV cutoff, so the displayed margins are not uniform in the field order. The finite claims are separate from our asymptotic theorem for each fixed field.

Related work. Exact input–output weight enumerators and decompositions by message weight also appear in analyses of turbo-like codes [BMS09]. Block–Accumulate codes apply this method to a block-diagonal outer map followed by recursive stages [KPR⁺26]. That work computes finite expected weight spectra and uses them to select concrete parameters. Its main code family changes the outer map. We instead retain the original EA and EC ensembles, replace their recursive-stage relaxations by exact transfers, and then measure the effect of regular expander incidence.

Recent large-field analyses of repeat–accumulate–accumulate (RAA) codes address a second source of slack: the union over many field assignments with the same set of nonzero message coordinates. Akhiani and Zhang group these assignments before the union bound and prove constant relative distance over sufficiently large fields [AZ26]. Khabbazi sharpens the analysis for randomly scaled RAA codes. The proof fixes the support geometry, identifies scalar multiples, and charges only the cancellation equations that exceed the remaining message degrees of freedom [Kha26].

Our support grouping in (112) uses the same accounting principle. The EA and EC ensembles require different structural counts. Expander collisions determine the rank of the accumulator constraints. For EC, the convolution state also determines which constraints are independent. Our transfer matrices therefore count output weight and independent equations together. The large-field RAA results study an asymptotic regime in which the field grows with the block length. Our near-128-bit headline results instead fix the field and length, then certify the complete finite union bound.

Organization. Section 2 defines the sampled maps and the distance event. Section 3 develops the common enumerator method. Sections 4 and 5 treat the original binary ensembles. Section 6 introduces regular expanders and gives the binary rate-one-half frontier. Section 7 gives the finite-field extension and separates fixed-field asymptotics from cryptographic-size finite bounds. Section 8 describes the certified evaluation, and Section 9 records limitations and open questions.

2 Code ensembles and distance events

This section defines the objects sampled in the paper. We use row vectors, so a message $\mathbf{x} \in \mathbb{F}_p^k$ is encoded as $\mathbf{x}\mathbf{G} \in \mathbb{F}_p^n$ by a matrix $\mathbf{G} \in \mathbb{F}_p^{k \times n}$. Throughout the paper, p denotes the order of the field. It may be a prime power; it does not necessarily denote the characteristic.

2.1 Distance and the GV benchmark

For $\mathbf{y} \in \mathbb{F}_p^n$, its support is $\text{supp}(\mathbf{y}) := \{i : y_i \neq 0\}$, and its Hamming weight is $\text{wt}(\mathbf{y}) := |\text{supp}(\mathbf{y})|$. Vectors of the same weight form a *Hamming shell*. Define

$$d_{\min}(\mathbf{G}) := \min_{\mathbf{x} \in \mathbb{F}_p^k \setminus \{0\}} \text{wt}(\mathbf{x}\mathbf{G}). \quad (1)$$

This convention assigns distance zero to a noninjective generator. If $d_{\min}(\mathbf{G}) > 0$, the matrix is injective and (1) equals the usual minimum distance of its image. The relative minimum distance is $d_{\min}(\mathbf{G})/n$.

For an integer cutoff L , define the bad-sampling event

$$\text{Bad}_L(\mathbf{G}) := \{d_{\min}(\mathbf{G}) \leq L\}. \quad (2)$$

Every probability in this paper is over the stated random choices used to sample $\mathbf{G}$. A bound $\Pr[\text{Bad}_L(\mathbf{G})] \leq U$ shows that a sampled generator has distance above L except with probability at most U .

The code rate is $R := k/n$. We use natural logarithms and define

$$h(x) := -x \ln x - (1-x) \ln(1-x), \quad (3)$$

$$h_p(x) := h(x) + x \ln(p-1). \quad (4)$$

The p -ary GV relative distance $\delta_{p,\text{GV}}(R)$ is the solution in $(0, (p-1)/p)$ to

$$h_p(\delta_{p,\text{GV}}(R)) = (1-R) \ln p. \quad (5)$$

This equation balances the number p^k of messages against the exponential volume of a radius- δn Hamming ball. We use it only as a distance benchmark. Here a radius- L Hamming ball centered at zero is the set of vectors of weight at most L ; it contains $\sum_{h=0}^L \binom{n}{h} (p-1)^h$ vectors.

2.2 The common architecture

Every construction has the form

$$\mathbf{G} = \mathbf{B}\mathbf{T}, \quad (6)$$

where $\mathbf{B} \in \mathbb{F}_p^{k \times n}$ is sparse and $\mathbf{T} \in \mathbb{F}_p^{n \times n}$ is an invertible recursive map. We call $\mathbf{B}$ the expander matrix, following the EA and EC literature. No spectral or combinatorial expansion property is implicit in that name.

The rows of $\mathbf{B}$ correspond to message coordinates. Its columns correspond to the intermediate coordinates supplied to $\mathbf{T}$. Thus the bipartite graph of $\mathbf{B}$ has k left vertices and n right vertices. A matrix entry is an edge when it is nonzero.

Accumulator. Given $\mathbf{u} \in \mathbb{F}_p^n$, initialize $y_0 := 0$ and compute

$$y_t := u_t + y_{t-1}, \quad t \in \{1, \dots, n\}. \quad (7)$$

The resulting linear map is the accumulator $\mathbf{A}$. It is invertible because $u_t = y_t - y_{t-1}$. The EA generator is $\mathbf{G} = \mathbf{B}\mathbf{A}$.

Wrapped binary convolution. Fix a memory $m \geq 1$. Initialize $y_j := 0$ for $j \leq 0$. For each time t , sample independent bits $b_{t,1}, \dots, b_{t,m-1}$ and compute

$$y_t := u_t + y_{t-m} + \sum_{j=1}^{m-1} b_{t,j} y_{t-j} \quad \text{over } \mathbb{F}_2. \quad (8)$$

The coefficient of the oldest state bit is fixed to one. This is the wrapped EC convolution of [RRT23]. Its coefficient on u_t is one, so every sampled convolution is invertible.

Table 2: Random choices in the principal ensembles.

name	field	expander	recursive map
binary EA	$\mathbb{F}_2$	Bernoulli	fixed accumulator
binary wrapped EC	$\mathbb{F}_2$	Bernoulli	independent wrapped convolution
regular binary EA	$\mathbb{F}_2$	left regular	fixed accumulator
regular binary EC	$\mathbb{F}_2$	left regular	independent wrapped convolution
labeled p -ary EA	$\mathbb{F}_p$	labeled Bernoulli or regular	fixed accumulator
labeled p -ary EC	$\mathbb{F}_p$	labeled two-sided regular	independent full-field convolution

Finite-field convolution. Fix a prime power p , choose the field $\mathbb{F}_p$, and fix memory $m \geq 1$. Initialize $y_j := 0$ for $j \leq 0$. At each time $t \in \{1, \dots, n\}$, independently sample $\alpha_t \leftarrow \mathbb{F}_p^m$ and compute

$$y_t := u_t + \langle \alpha_t, (y_{t-1}, \dots, y_{t-m}) \rangle. \quad (9)$$

This map is also invertible. Full-field coefficients give the uniform inner-product law used in Section 7.

2.3 Expander distributions

We use three principal distributions for $\mathbf{B}$.

Bernoulli incidence. Over $\mathbb{F}_2$, independently set each entry of $\mathbf{B}$ to one with probability ρ and to zero otherwise. We write $\mathbf{B} \leftarrow \mathcal{B}_{k,n,\rho}^{\text{Ber}}$. A row has expected weight ρn .

Over $\mathbb{F}_p$, an entry is zero with probability $1 - \rho$. Otherwise it is sampled from $\mathbb{F}_p^*$. We call the nonzero value an edge label. All incidence and label choices are independent unless stated otherwise.

Region-stratified left regularity. Assume $n = d_L \ell$ and partition the right coordinates into d_L consecutive regions of length ℓ . For each left vertex and each region, choose one right coordinate uniformly and independently. Place one edge there. Over $\mathbb{F}_p$, independently label the edge with an element of $\mathbb{F}_p^*$.

Every left vertex has degree d_L , with one edge in every region. Right degrees remain random. We call this distribution a *region-stratified left-regular expander*. We use *regular expander* when the one-sided meaning is already clear.

Two-sided regularity. Assume $k = d_R \ell$ and $n = d_L \ell$, so $kd_L = nd_R$. Within each region, sample a permutation of the k left vertices. Split the permuted list into ℓ groups of size d_R and connect each group to one right coordinate. Every left vertex has one edge per region, and every right vertex has degree d_R . We call this a *two-sided regular expander*.

At rate one half, $d_R = d_L/2$. The finite-field EC certificates in Section 8 use this distribution. The binary regular EA certificate uses left regularity. The binary EC frontier includes both left-regular and two-sided regular ensembles.

The word *independent* in Table 2 means that the expander and recursive-map randomness are sampled independently. Within the regular expander, incidences have the dependencies imposed by the stated degree constraints.

Sampling and cost. The following procedures make the setup cost explicit. For a left-regular expander, process regions $j = 1, \dots, d_L$. For every left vertex i , sample $c_{i,j} \leftarrow \{1, \dots, \ell\}$ and connect i to coordinate $(j-1)\ell + c_{i,j}$. For a two-sided regular expander, sample a permutation π_j of the k

left vertices in each region. Connect $\pi_j((q-1)d_R+1), \dots, \pi_j(qd_R)$ to the q th coordinate of that region. Over $\mathbb{F}_p$, sample one independent label from $\mathbb{F}_p^*$ for every generated edge.

Both procedures generate exactly kd_L edges; in the two-sided ensemble, this also equals nd_R . A materialized encoder stores that many indices and, over $\mathbb{F}_p$, the same number of labels. It uses one scatter-add per binary edge or one multiply-add per labeled edge. Left-regular setup samples kd_L independent indices. Two-sided setup uses d_L permutations of length k ; a Fisher–Yates implementation takes $O(kd_L)$ random words and $O(k)$ reusable workspace. A streaming encoder may regenerate one region at a time from a public seed and reduce storage to $O(k)$ indices. The certificate comparisons count edges only. They do not claim that permutations, transposition, and field multiplications have equal cost.

Encoding with a stored sparse expander uses $O(kd_L+n)$ field operations for EA and $O(kd_L+nm)$ for EC. For Bernoulli incidence, the expected number of edges is $kn\rho$. At fixed rate and fixed memory, expected encoding cost is therefore $O(n \log n)$ when $n\rho = C \ln n$ with fixed C , and $O(n \log^2 n)$ when $n\rho = (\ln n)^2$. Theorem 7.5 approaches GV in the first regime with tunable polynomial sampling failure. Corollary 7.6 uses the second regime to make sampling failure negligible.

Parallel depth of EA. Depth counts the longest chain of dependent field operations, with each addition or multiplication taking one step and each operation having at most two inputs. Fix the sampled matrix and its sparse layout before encoding; this arithmetic model does not charge for setup or data movement. Let E be the number of nonzero entries of $\mathbf{B}$, and let Δ_R be the largest number of nonzero entries in a column. Independent edge products followed by balanced sums compute $\mathbf{x}\mathbf{B}$ in $O(E+n)$ work and $O(1+\log(1+\Delta_R))$ depth. The accumulator is a prefix sum, which has $O(n)$ work and $O(\log n)$ depth [Ble90]. Thus EA encoding has $O(E+n)$ work and $O(1+\log(1+\Delta_R)+\log n)$ depth.

Only one prefix sum is needed. In the usual staged evaluation of RAA, the first accumulation feeds a permutation and then a second accumulation. Each additional accumulator adds another dependent scan, although a fixed number of stages still gives $O(\log n)$ arithmetic depth. This is a comparison of the natural staged encoders, not a circuit-depth lower bound for RAA. Our GV result for EA does not require adding such stages.

The same distinction applies to transposed encoding. For a column vector $\mathbf{v}$, compute $\mathbf{G}\mathbf{v} = \mathbf{B}(\mathbf{A}\mathbf{v})$: first one suffix sum, then the sparse map. Here $\mathbf{A}$ is upper triangular under our row-vector convention, so its action on a column vector is a suffix sum rather than a prefix sum. The depth bound replaces Δ_R by the largest row weight of $\mathbf{B}$. This full-vector computation is separate from the local PCF evaluation, which uses comparison functions to avoid scanning the full noise vector.

3 The enumerator method

The minimum-distance event in (2) quantifies over all nonzero messages. We first count bad messages in expectation. Exact enumerators make the expectation computable without replacing the recursive state process by a concentration inequality.

3.1 Input–output weight enumerators

Let $\mathcal{D}$ be a distribution on matrices in $\mathbb{F}_p^{a \times b}$. Its expected input–output weight enumerator is

$$E_{r,u}^{\mathcal{D}} := \mathbb{E}_{\mathbf{M} \leftarrow \mathcal{D}} \left| \left\{ \mathbf{x} \in \mathbb{F}_p^a : \text{wt}(\mathbf{x}) = r, \text{wt}(\mathbf{x}\mathbf{M}) = u \right\} \right|. \quad (10)$$

This is an expectation over matrices, not a count for one fixed matrix.

For a sampled generator $\mathbf{G}$, define

$$Z_L(\mathbf{G}) := \left| \left\{ \mathbf{x} \in \mathbb{F}_p^k \setminus \{0\} : \text{wt}(\mathbf{x}\mathbf{G}) \leq L \right\} \right|. \quad (11)$$

The event $\text{Bad}_L(\mathbf{G})$ implies $Z_L(\mathbf{G}) \geq 1$. Therefore

$$\Pr[\text{Bad}_L(\mathbf{G})] \leq \mathbb{E}[Z_L(\mathbf{G})]. \quad (12)$$

This is the only use of Markov's inequality in the common argument. Our improvement concerns how $\mathbb{E}[Z_L]$ is evaluated.

3.2 Composition under exchangeability

A monomial matrix has one nonzero entry in every row and column. It permutes coordinates and scales them by nonzero field elements. A random vector whose law is invariant under every monomial matrix is uniform on each Hamming shell after conditioning on its weight.

Lemma 3.1 (Exchangeable serial composition). *Let $\mathbf{B} \in \mathbb{F}_p^{k \times n}$ be sampled so that $\mathbf{B}\Pi$ and $\mathbf{B}$ have the same distribution for every monomial matrix $\Pi \in \mathbb{F}_p^{n \times n}$. Independently sample $\mathbf{T} \in \mathbb{F}_p^{n \times N}$. Then*

$$E_{r,h}^{\mathbf{B}\mathbf{T}} = \sum_{u=0}^n \frac{E_{r,u}^{\mathbf{B}} E_{u,h}^{\mathbf{T}}}{\binom{n}{u} (p-1)^u}. \quad (13)$$

Proof. Fix a message $\mathbf{x}$ and condition on $\text{wt}(\mathbf{x}\mathbf{B}) = u$. Monomial invariance makes $\mathbf{x}\mathbf{B}$ uniform on the p -ary weight- u shell. Averaging the independent map $\mathbf{T}$ over that shell gives the second enumerator divided by the shell size $\binom{n}{u} (p-1)^u$. Summing over u and then over weight- r messages proves the identity. $\square$

For binary matrices, coordinate-permutation symmetry alone suffices. The Bernoulli and global fixed-row ensembles satisfy the lemma. A region-stratified expander is exchangeable only within each region. For that ensemble we retain the recursive state at regional boundaries instead of using (13) globally.

3.3 Transfer matrices

Suppose a recursive map has a finite state set $\mathcal{S}$. Associate a monomial $X^a Y^b$ with a transition that consumes input weight a and emits output weight b . Let $\mathbf{T}(X, Y)$ contain the transition probabilities or expected transition counts. Assume that, conditional on the current state, this weighted transition law is the same at every step and does not depend on the earlier path. Fresh independent randomness at each step ensures this property for the recursive maps below. If $\mathbf{e}_0$ selects the initial state, then

$$[X^u Y^h] \mathbf{e}_0^\top \mathbf{T}(X, Y)^n \mathbf{1} \quad (14)$$

is the expected mass of length- n paths with input weight u and output weight h . Matrix multiplication preserves the order of the recursive process, while coefficient extraction forgets everything except the two weights.

For a one-variable probability generating matrix $\mathbf{M}(Y)$ with nonnegative coefficients, define

$$F(Y) := \mathbf{e}_0^\top \mathbf{M}(Y)^n \mathbf{1}.$$

Every numerical marker $z \in (0, 1]$ gives

$$\sum_{h=0}^L [Y^h] F(Y) \leq z^{-L} F(z). \quad (15)$$

The left side uses coefficient notation schematically: the matrix on the right is evaluated at the fixed marker z . The inequality follows because $z^h \geq z^L$ for $h \leq L$. Optimizing z improves the numerical bound, but any fixed valid marker gives a sound bound.

3.4 Covering all message weights

A finite certificate must cover every $1 \leq r \leq k$. Evaluating selected weights is not sufficient. Our verifiers use three regimes:

1. exact transfer or shell recurrences for small supports;
2. one positive-marker bound for each contiguous support interval;
3. a point-mass or Hamming-ball bound for dense supports.

Every certificate records an explicit partition of $[1, k]$. Its verifier rejects overlapping intervals, gaps, invalid markers, or a final sum above the claimed value. Section 8 describes the trusted arithmetic.

The following elementary lemma isolates the common block calculation. The theorem-specific analysis supplies the envelope and proves its shape. The verifier only evaluates the stated endpoints and checks the partition.

Lemma 3.2 (Certified support partition). *Let $f_r \geq 0$ be the first-moment contribution of support r , and let $\mathcal{I}$ be a disjoint partition of $\{1, \dots, k\}$ into integer intervals. For each $I = [a, b] \in \mathcal{I}$, suppose a fixed valid marker defines an upper envelope $F_I : [a, b] \rightarrow \mathbb{R}_{\geq 0}$ with $f_r \leq F_I(r)$ for every integer $r \in I$. Define*

$$M_I := \begin{cases} F_I(b), & F_I \text{ is nondecreasing,} \\ F_I(a), & F_I \text{ is nonincreasing,} \\ F_I(a) + F_I(b), & F_I > 0 \text{ and } \ln F_I \text{ is convex.} \end{cases} \quad (16)$$

Any applicable line gives

$$\sum_{r=1}^k f_r \leq \sum_{I=[a,b] \in \mathcal{I}} (b - a + 1) M_I. \quad (17)$$

The same conclusion holds when a theorem supplies a direct upper bound $B_I \geq \sum_{r \in I} f_r$, with B_I replacing $(b - a + 1) M_I$.

Proof. Monotonicity gives the first two endpoint bounds. If $\ln F_I$ is convex, its maximum on the real interval $[a, b]$ occurs at an endpoint. Thus $F_I(r) \leq \max\{F_I(a), F_I(b)\} \leq F_I(a) + F_I(b)$. Summing the pointwise bound over each interval proves (17). The direct-bound case follows by summing the inequalities $\sum_{r \in I} f_r \leq B_I$. $\square$

An interval verifier implements this lemma with outward-rounded enclosures. It checks that the intervals partition $[1, k]$, that each marker is in its proved domain, and that the upper endpoint of every computed M_I or B_I is finite. Symbolic monotonicity and convexity remain mathematical claims; the verifier invokes them only through the named bounds identified in Section 8.

4 Binary Expand–Accumulate

We first keep the binary EA construction of [BCG⁺22] unchanged. The only change is the analysis. For each message weight, the Bernoulli expander gives independent intermediate bits. The accumulator then has an exact two-state description.

4.1 Exact first moment

Fix a message $\mathbf{x} \in \mathbb{F}_2^k$ of weight r and sample $\mathbf{B} \leftarrow \mathcal{B}_{k,n,\rho}^{\text{Ber}}$. Each coordinate of $\mathbf{x}\mathbf{B}$ is the parity of r independent Bernoulli variables. Its probability of being one is

$$a_r := \frac{1 - (1 - 2\rho)^r}{2}. \quad (18)$$

Different coordinates use independent columns of $\mathbf{B}$. Thus the entries of $\mathbf{x}\mathbf{B}$ are independent $\text{Ber}(a_r)$ variables.

Let Y mark a nonzero accumulator output. Rows and columns of

$$\mathbf{T}_a(Y) := \begin{pmatrix} 1 - a & aY \\ a & (1 - a)Y \end{pmatrix} \quad (19)$$

index the current and next accumulator states in the order 0, 1. The factor Y marks the next state because that state is also the current output symbol. Let $\mathbf{e}_0 := (1, 0)^\top$.

Theorem 4.1 (Exact Bernoulli EA first moment). *For $\mathbf{G} = \mathbf{B}\mathbf{A}$ and every $0 \leq L \leq n$, define*

$$\mu_{\text{EA}}(L) := \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L [Y^h] \mathbf{e}_0^\top \mathbf{T}_{a_r}(Y)^n \mathbf{1}. \quad (20)$$

Then

$$\Pr_{\mathbf{B}}[\text{Bad}_L(\mathbf{B}\mathbf{A})] \leq \mu_{\text{EA}}(L). \quad (21)$$

Proof. For a fixed weight- r message, the matrix product in (20) is the probability generating function of its output weight. Coefficient extraction sums its probability of output weight at most L . There are $\binom{k}{r}$ binary messages of weight r . Equation (12) gives the probability bound. $\square$

The theorem retains the exact accumulator process. The next bound replaces coefficient extraction by a one-dimensional optimization without changing the transfer matrix.

For $a \in [0, 1/2]$ and $z \in (0, 1]$, define

$$\lambda(a, z) := \frac{(1 - a)(1 + z) + \sqrt{(1 - a)^2(1 - z)^2 + 4a^2z}}{2}, \quad (22)$$

$$J_\delta(a) := \sup_{0 < z \leq 1} \{\delta \ln z - \ln \lambda(a, z)\}. \quad (23)$$

The quantity $\lambda(a, z)$ is the spectral radius of $\mathbf{T}_a(z)$, meaning the largest absolute value of its eigenvalues.

Lemma 4.2 (Accumulator lower tail). *Let H be the output weight of the binary accumulator on n independent $\text{Ber}(a)$ inputs. For $a \in (0, 1/2]$ and $\delta \in (0, 1/2)$,*

$$\Pr[H \leq \lfloor \delta n \rfloor] \leq \sqrt{2} \exp(-nJ_\delta(a)). \quad (24)$$

Moreover,

$$\liminf_{a \downarrow 0} \frac{J_\delta(a)}{a} \geq I_\delta := 1 - 2\sqrt{\delta(1-\delta)}, \quad (25)$$

$$J_\delta(1/2) = \ln 2 - h(\delta). \quad (26)$$

The function J_δ is continuous and positive on $(0, 1/2]$.

Proof. Set $D_z := \text{diag}(1, \sqrt{z})$. The similar matrix

$$D_z \mathbf{T}_a(z) D_z^{-1} = \begin{pmatrix} 1-a & a\sqrt{z} \\ a\sqrt{z} & (1-a)z \end{pmatrix}$$

is symmetric and positive semidefinite. Its largest eigenvalue is $\lambda(a, z)$. Hence

$$\mathbb{E}[z^H] = \mathbf{e}_0^\top \mathbf{T}_a(z)^n \mathbf{1} \leq \sqrt{1+z} \lambda(a, z)^n.$$

Apply (15) and optimize over z .

For the sparse endpoint, set $z = e^{-\theta a}$ and expand at $a = 0$. The coefficient of a is optimized at $\theta = (1 - 2\delta)/\sqrt{\delta(1-\delta)}$, which gives (25). At $a = 1/2$, one has $\lambda(1/2, z) = (1+z)/2$. The choice $z = \delta/(1-\delta)$ gives (26). For $a > 0$, the derivative of $\ln \lambda(a, z)$ with respect to $\ln z$ at $z = 1$ is the stationary nonzero-state probability $1/2 > \delta$. A marker slightly below one therefore gives a positive exponent. Also, $\lambda(a, z) \geq 1 - a \geq 1/2$, so the defining objective is at most $\delta \ln z + \ln 2$. This tends uniformly to $-\infty$ as $z \downarrow 0$. Restricting the optimization to a compact interval of positive markers proves continuity. $\square$

4.2 Asymptotic distance

The endpoint formulas separate the sparse and dense message regimes. Sparse messages determine the required expander density. Dense messages give the GV condition.

Theorem 4.3 (Binary EA distance). *Let $k = k(n)$ satisfy $k/n \rightarrow R \in (0, 1)$. Sample $\mathbf{B} \leftarrow \mathcal{B}_{k,n,\rho_n}^{\text{Ber}}$ with*

$$\rho_n = C \frac{\ln n}{n} \quad (27)$$

for a constant $C > 0$. Fix $\delta \in (0, 1/2)$. If

$$CI_\delta > 1, \quad (28)$$

$$h(\delta) < (1 - R) \ln 2, \quad (29)$$

then

$$\Pr_{\mathbf{B}}[d_{\min}(\mathbf{B}\mathbf{A}) \leq \lfloor \delta n \rfloor] = o(1). \quad (30)$$

Proof. We apply Lemma 4.2 to Theorem 4.1 and split the message weights into three ranges.

Choose constants $\varepsilon, \tau > 0$ and $\gamma > 1$ such that $C(1-\tau)(I_\delta - \varepsilon) > \gamma$. Decrease τ until $J_\delta(a) \geq a(I_\delta - \varepsilon)$ for $0 < a \leq \tau$, as permitted by (25). For all sufficiently large n and every $r\rho_n \leq \tau$,

$$r\rho_n(1 - r\rho_n) \leq \frac{1 - e^{-2r\rho_n}}{2} \leq a_r \leq r\rho_n.$$

Lemma 4.2 therefore bounds the weight- r tail by $\sqrt{2} n^{-\gamma r}$. Since $\binom{k}{r} \leq k^r$ and $k = O(n)$, summing this geometric bound over the sparse range gives $o(1)$.

For $r\rho_n > \tau$, one has $a_r \geq (1 - e^{-2\tau})/2$. Positivity and continuity of J_δ give a uniform exponential tail. The number of messages with $r \leq \eta n$ has exponent $Rh(\eta/R) + o(1)$. Choose $\eta > 0$ small enough that this counting exponent is below the tail exponent. The intermediate contribution is then exponentially small.

For $r > \eta n$, equation (18) gives $a_r = 1/2 - o(1)$ uniformly. There are at most 2^k messages. The dense endpoint bounds their total contribution by

$$\exp(n(R \ln 2 + h(\delta) - \ln 2 + o(1))),$$

which vanishes by (29). The three ranges cover every nonzero message weight. $\square$

Condition (29) is exactly the binary GV condition. The accumulator therefore loses no asymptotic exponent for dense messages. The sparse condition is an additional cost of the sparse expander.

At $\delta = 0.05$, the sparse condition requires $C > 1.773$. The analysis in [BCG⁺22, Theorem 3.10] requires $C > 4.938$. At rate $R = 1/5$, its rate condition permits $\delta < 0.1127$, while (29) permits $\delta < 0.2430$. These are asymptotic comparisons; they do not state a finite failure probability.

4.3 Finite same-ensemble certificates

The verifier described in Section 8 evaluates (20) over every message weight. Its block bound uses the following monotonicity argument. Write $S_a(z) := D_z \mathbf{T}_a(z) D_z^{-1}$ for the symmetric matrix in Lemma 4.2. Its derivative is

$$\frac{\partial S_a(z)}{\partial a} = - \begin{pmatrix} 1 \\ -\sqrt{z} \end{pmatrix} \begin{pmatrix} 1 & -\sqrt{z} \end{pmatrix}.$$

Every quadratic form of this derivative is nonpositive. Thus the largest eigenvalue $\lambda(a, z)$ is nonincreasing in a . For $0 < \rho < 1/2$, the activation probability a_r is nondecreasing in r . Consequently, for any integer interval $[u, v] \subseteq [1, k/2]$ and $0 < z \leq 1$,

$$\sum_{r=u}^v \binom{k}{r} \Pr[H_r \leq L] \leq (v - u + 1) e^{kh(v/k)} \sqrt{1+z} z^{-L} \lambda(a_u, z)^n, \quad (31)$$

where H_r is the accumulator-output weight for a fixed weight- r message. Here $\binom{k}{r} \leq e^{kh(r/k)}$, and h is nondecreasing on $[0, 1/2]$. For a block above $k/2$, replace the count factor $(v - u + 1) e^{kh(v/k)}$ by 2^k .

Theorem 4.4 (Finite Bernoulli EA certificates). *For each row of Table 3, set $k := 2^s$, $n := 5k$, $\rho := w/n$, and $L := n/20$. If $\mathbf{B} \leftarrow \mathcal{B}_{k,n,\rho}^{\text{Ber}}$, then*

$$\Pr_{\mathbf{B}}[\text{Bad}_L(\mathbf{BA})] \leq U < 2^{-20}. \quad (32)$$

Proof. For each row, the verifier evaluates weights 1 through 128 with the exact transfer matrix. It partitions the remaining weights into contiguous intervals and applies (31) or its 2^k count variant. Their union is $[129, k]$. Outward-rounded interval arithmetic evaluates every stored marker and the final sum. The verified upper endpoint is the displayed value U . Theorem 4.1 converts that sum into the claimed probability bound. $\square$

At $s = 20$, the original $C = 3$ profile has expected row weight 46.42 but an extrapolated failure bound 0.0157 at distance 0.05 [BCG⁺22]. The first row of Table 3 is slightly denser and proves a complete 2^{-20} statement. At $s = 25$ and $s = 30$, the certified profiles are both sparser and have much smaller failure bounds than the corresponding published profiles.

Table 3: Certified binary Bernoulli EA bounds at rate $1/5$ and relative distance 0.05 . The parameter w is the expected expander row weight.

s	w	U	$-\log_2 U$
20	50	$6.891818499 \cdot 10^{-7}$	20.468611
25	56	$7.475946766 \cdot 10^{-7}$	20.351240
30	62	$8.107513660 \cdot 10^{-7}$	20.234237

5 Binary Expand–Convolute

The accumulator has one state bit. A random convolution has m state bits, but its output-weight law does not require all 2^m bit patterns. It is enough to record how many consecutive zeros end the current output prefix. This reduction gives an exact transfer matrix with $m + 1$ states.

We analyze the wrapped convolution from (8). The fixed oldest tap forces the process to leave the almost-zero state unless the next input cancels it. This behavior improves the sparse-message exponent.

5.1 Exact transfer matrix

Index the reduced states by $0, \dots, m$. State $j < m$ means that the current output prefix ends in exactly j zeros. State m means that its last m outputs are all zero. The initial state is m .

Let X mark nonzero input bits and let Y mark nonzero output bits. Define the $(m + 1) \times (m + 1)$ matrix $\mathbf{W}_m(X, Y)$ by its nonzero entries:

$$(\mathbf{W}_m)_{j,0} = \frac{1+X}{2}Y, \quad (\mathbf{W}_m)_{j,j+1} = \frac{1+X}{2} \quad (0 \leq j < m-1), \quad (33)$$

$$(\mathbf{W}_m)_{m-1,0} = Y, \quad (\mathbf{W}_m)_{m-1,m} = X, \quad (34)$$

$$(\mathbf{W}_m)_{m,0} = XY, \quad (\mathbf{W}_m)_{m,m} = 1. \quad (35)$$

Lemma 5.1 (Exact wrapped-convolution enumerator). *Let $E_{u,h}^{\text{wConv}}$ be the expected number of weight- u inputs that a sampled wrapped convolution maps to weight h . Then*

$$E_{u,h}^{\text{wConv}} = [X^u Y^h] \mathbf{e}_m^T \mathbf{W}_m(X, Y)^n \mathbf{1}, \quad (36)$$

where $\mathbf{e}_m$ selects the all-zero initial state.

Proof. If $j < m - 1$, at least one state bit with a random tap is nonzero. Its inner product with the sampled taps is uniform, independently of the input bit. This gives (33). In state $m - 1$, only the fixed oldest tap is active. Input zero produces output one; input one produces zero and enters state m . This gives (34). In state m , the feedback is zero and the output equals the input. Matrix multiplication concatenates the transitions. $\square$

If the input bits are independent $\text{Ber}(a)$ variables, averaging the input marker gives a one-variable matrix $\mathbf{N}_{m,a}(Y)$ with nonzero entries

$$(\mathbf{N}_{m,a})_{j,0} = Y/2, \quad (\mathbf{N}_{m,a})_{j,j+1} = 1/2 \quad (0 \leq j < m-1), \quad (37)$$

$$(\mathbf{N}_{m,a})_{m-1,0} = (1-a)Y, \quad (\mathbf{N}_{m,a})_{m-1,m} = a, \quad (38)$$

$$(\mathbf{N}_{m,a})_{m,0} = aY, \quad (\mathbf{N}_{m,a})_{m,m} = 1-a. \quad (39)$$

Theorem 5.2 (Exact Bernoulli wrapped-EC first moment). *Independently sample $\mathbf{B} \leftarrow \mathcal{B}_{k,n,\rho}^{\text{Ber}}$ and a wrapped convolution $\mathbf{C}$ of memory m . Let a_r be defined by (18). Then*

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L [Y^h] \mathbf{e}_m^\top \mathbf{N}_{m,a_r}(Y)^n \mathbf{1}. \quad (40)$$

Proof. For a fixed weight- r message, the Bernoulli expander produces independent $\text{Ber}(a_r)$ inputs to the convolution. Matrix $\mathbf{N}_{m,a_r}$ gives the exact output-weight generating function. Sum over output weights and messages, then apply (12). $\square$

For a numerical marker $z \in (0, 1]$, the inner tail in (40) is at most

$$z^{-L} \mathbf{e}_m^\top \mathbf{N}_{m,a_r}(z)^n \mathbf{1}. \quad (41)$$

The calculation uses $m + 1$ states and remains exact before the positive-marker bound.

To cover an interval of message weights, the verifier needs one transfer bound valid throughout an interval of activation probabilities. For $0 \leq a_- \leq a \leq a_+ < 1$, coefficient positivity gives the entrywise inequality

$$\mathbf{N}_{m,a}(z) = (1 - a) \mathbf{W}_m\left(\frac{a}{1 - a}, z\right) \leq (1 - a_-) \mathbf{W}_m\left(\frac{a_+}{1 - a_+}, z\right) =: \mathbf{M}_{m;a_-,a_+}(z). \quad (42)$$

All entries are nonnegative, so this inequality is preserved by matrix powers. Substituting $\mathbf{M}_{m;a_-,a_+}(z)^n$ in (41) gives a uniform tail bound. For a support interval $[u, v]$, take $a_- := a_u$ and $a_+ := a_v$. Multiply the tail bound by an upper bound on $\sum_{r=u}^v \binom{k}{r}$ to obtain the interval contribution.

For dense supports, a simpler bound holds for every fixed invertible convolution. When $a_r \leq 1/2$, each particular expander output has probability at most $(1 - a_r)^n$. Invertibility gives one preimage for each output in the Hamming ball, hence

$$\Pr[\text{wt}(\mathbf{xBC}) \leq L] \leq \left(\sum_{h=0}^L \binom{n}{h} \right) (1 - a_r)^n. \quad (43)$$

The right side is nonincreasing in r , so its value at the left endpoint bounds a dense support interval after multiplying by the message count.

5.2 Asymptotic distance

Let $\varrho_m(a, z)$ be the spectral radius of $\mathbf{N}_{m,a}(z)$, namely the largest absolute value of its eigenvalues. Define

$$J_{m,\delta}^w(a) := \sup_{0 < z \leq 1} \{\delta \ln z - \ln \varrho_m(a, z)\}, \quad (44)$$

$$I_{m,\delta}^w := \left(\sqrt{1 - 2\delta + 2^{1-m}\delta} - \sqrt{2^{1-m}\delta} \right)^2. \quad (45)$$

Lemma 5.3 (Wrapped-convolution endpoint exponents). *For fixed $m \geq 1$ and $\delta \in (0, 1/2)$, $J_{m,\delta}^w$ is continuous and positive on $(0, 1/2]$. Moreover,*

$$\liminf_{a \downarrow 0} \frac{J_{m,\delta}^w(a)}{a} \geq I_{m,\delta}^w, \quad (46)$$

$$J_{m,\delta}^w(1/2) = \ln 2 - h(\delta). \quad (47)$$

The associated lower-tail bounds can be chosen uniformly for a near zero and for a in any closed interval bounded away from zero.

Proof. Perron–Frobenius theory bounds $\mathbf{e}_m^\top \mathbf{N}_{m,a}(z)^n \mathbf{1}$ by a constant times $\varrho_m(a, z)^n$. At $a = 0$ and $z = 1$, the chain has an active class and an all-zero class. First-order perturbation with $z = e^{-\theta a}$ reduces the calculation to a two-state continuous-time generator. Optimizing its larger eigenvalue gives (45). At $a = 1/2$, every row of $\mathbf{N}_{m,1/2}(z)$ sums to $(1+z)/2$. Hence $\varrho_m(1/2, z) = (1+z)/2$, which gives (47). Appendix C gives the perturbation and proves a uniform quadratic remainder in Lemma C.1. It also proves the uniform prefactor claim. $\square$

Theorem 5.4 (Binary wrapped-EC distance). *Fix $m \geq 1$ and $\delta \in (0, 1/2)$. Let $k/n \rightarrow R \in (0, 1)$ and set $\rho_n = C \ln(n)/n$. Independently sample $\mathbf{B} \leftarrow \mathcal{B}_{k,n,\rho_n}^{\text{Ber}}$ and a wrapped convolution $\mathbf{C}$ of memory m . If*

$$CI_{m,\delta}^w > 1, \quad (48)$$

$$h(\delta) < (1 - R) \ln 2, \quad (49)$$

then

$$\Pr_{\mathbf{B}, \mathbf{C}} [d_{\min}(\mathbf{BC}) \leq \lfloor \delta n \rfloor] = o(1). \quad (50)$$

Proof. Apply Lemma 5.3 to the exact first moment in Theorem 5.2. Choose fixed thresholds $\varepsilon, \tau > 0$ and $\gamma > 1$ as in the proof of Theorem 4.3, replacing I_δ by $I_{m,\delta}^w$. The uniform sparse tail in (161) then bounds each weight- r tail by $Kn^{-\gamma r}$ for $r\rho_n \leq \tau$, with K independent of n, r . This makes the sparse contribution summable. For intermediate weights, a_r is bounded away from zero, so a uniform exponential tail dominates the number of messages. For every $r > \eta n$, one has $a_r = 1/2 - o(1)$ uniformly. The dense marker $z = \delta/(1 - \delta)$, its uniform Perron prefactor, and the bound of 2^k messages give exponent

$$R \ln 2 + h(\delta) - \ln 2 + o(1),$$

which is negative by (49). These three ranges cover $[1, k]$. $\square$

For $m = 1$, equation (45) equals the EA value I_δ . As $m \rightarrow \infty$, it tends to $1 - 2\delta$. At $\delta = 0.05$ and $m = 21$, the new sparse condition is $C > 1.111623$. The published nonwrapping theorem uses $m = C_m \log_2 n$ with $C_m > 1$ and imposes $C > 10.4344$ and $R < 0.1278$ at the same distance [RRT23, Theorem 3]. These are two of its parameter restrictions, not a sufficient condition for its failure bound to vanish; that bound also depends on C_m . Our fixed-memory wrapped theorem permits every $R < 0.7136$, the binary GV rate at distance 0.05.

The asymptotic comparison changes the analyzed member of the EC family: the published theorem is nonwrapping, while Theorem 5.4 is for the wrapped construction already proposed in [RRT23]. The next comparison holds the wrapped finite ensemble fixed.

5.3 Finite same-ensemble comparison

Set $k := 2^{20}$, $n := 5k$, $L := n/20$, and $m := 5$. For each row of Table 4, set $\rho := C_0 \ln(n)/n$. The boundary state and expander distribution match the zero-state wrapped rows of [RRT23, Table 4].

Theorem 5.5 (Finite wrapped-EC comparison). *For each row, independently sample $\mathbf{B} \leftarrow \mathcal{B}_{k,n,\rho}^{\text{Ber}}$ and a wrapped convolution $\mathbf{C}$ of memory 5. The exact-transfer column of Table 4 upper-bounds $\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})]$.*

Table 4: Same-ensemble comparison with the prior wrapped-EC calculation.

C_0	expected degree	prior bound	exact transfer	failure exponent
3.0	46.4171	$2.26 \cdot 10^{-7}$	$2.187816572 \cdot 10^{-10}$	32.089789
2.5	38.6810	$1.72 \cdot 10^{-5}$	$9.485867880 \cdot 10^{-8}$	23.329645
2.3	35.5865	$1.73 \cdot 10^{-4}$	$1.124903039 \cdot 10^{-6}$	19.761768

Proof. The verifier evaluates weights 1 through 32 individually with (41). It covers the remaining weights below $k/4$ with contiguous transfer blocks. From $k/4$ onward, invertibility bounds each preimage by the output Hamming ball. The verified sums are the displayed exact-transfer values. Theorem 5.2 gives the probability statement. $\square$

The improvement factors are at least 1032, 181, and 153. The prior finite calculation assumes an extension of its comparison lemma. The exact transfer calculation does not use that assumption.

6 Regular expander variants

Sections 4 and 5 separate the distance proof into sparse, intermediate, and dense message weights. The dense regime already attains the GV exponent. The remaining expander cost comes from small supports. Region-stratified regularity reduces the chance that their edges cluster in a small part of the output.

6.1 The regional shell law

Assume $n = d_L \ell$ and sample the left-regular expander from Section 2. Fix a binary message of weight r . Within one region, its image is the parity of r independent uniform unit vectors in $\mathbb{F}_2^\ell$. A unit vector has one coordinate equal to one and all other coordinates equal to zero.

Let $\pi_{r,u}^{(\ell)}$ be the probability that this regional image has weight u . Initialize $\pi_{0,0}^{(\ell)} := 1$. Terms outside $\{0, \dots, \ell\}$ are zero, and

$$\pi_{r+1,v}^{(\ell)} = \pi_{r,v-1}^{(\ell)} \frac{\ell - v + 1}{\ell} + \pi_{r,v+1}^{(\ell)} \frac{v + 1}{\ell}. \quad (51)$$

Lemma 6.1 (Regional shell law). *For a fixed weight- r message, the d_L regional images are independent. Their weights have distribution $\pi_{r,\cdot}^{(\ell)}$. Conditional on weight u , a regional image is uniform among the binary vectors of weight u .*

Proof. If the current regional image has weight u , the next unit vector decreases its weight with probability u/ℓ and increases it otherwise. This proves (51). Coordinate-permutation symmetry gives conditional uniformity. Incidences in distinct regions are independent. $\square$

The recurrence is exact and contains only nonnegative quantities. It is practical for small message supports. Two comparison bounds cover larger supports. Define

$$a_r^{\text{Reg}} := \frac{1 - e^{-2r/\ell}}{2}, \quad \gamma_r := \frac{e^r r!}{r^r}. \quad (52)$$

Lemma 6.2 (Poisson comparison). *Let $\mathbf{Z} \in \mathbb{F}_2^n$ have independent $\text{Ber}(a_r^{\text{Reg}})$ coordinates. For every nonnegative function $F : \mathbb{F}_2^n \rightarrow \mathbb{R}_{\geq 0}$,*

$$\mathbb{E}_{\mathbf{B}}[F(\mathbf{xB})] \leq \gamma_r^{d_L} \mathbb{E}[F(\mathbf{Z})]. \quad (53)$$

Proof. In each region, replace the r unit-vector samples by independent Poisson counts of mean r/ℓ . Conditional on total count r , the counts have the original multinomial law. Without conditioning, their parities are independent $\text{Ber}(a_r^{\text{Reg}})$ variables. A Poisson variable of mean r equals r with probability γ_r^{-1} . Dropping the independent conditioning event in all d_L regions proves the claim. $\square$

Stirling's formula gives $\gamma_r = \Theta(\sqrt{r})$. Thus the comparison cost is polynomial in r when d_L is fixed. The cost is also nondecreasing for positive integer r , since $\gamma_{r+1}/\gamma_r = e(1 + 1/r)^{-r} > 1$.

The next lemma bounds the probability of each particular expanded vector, rather than only its weight. We call such a probability a point mass.

Lemma 6.3 (Regular point-mass bound). *For every fixed weight- r message and every $\mathbf{y} \in \mathbb{F}_2^n$,*

$$\Pr_{\mathbf{B}}[\mathbf{x}\mathbf{B} = \mathbf{y}] \leq 2^{d_L - n} \left(1 + e^{-2r/\ell}\right)^n. \quad (54)$$

Proof. For one region, test a vector by the parity of a fixed set of j coordinates. One uniform unit step preserves this parity minus the probability that it flips it, giving average $1 - 2j/\ell$; after r independent steps the average is $(1 - 2j/\ell)^r$. Fourier inversion and the triangle inequality bound a regional point probability by

$$2^{-\ell} \sum_{j=0}^{\ell} \binom{\ell}{j} |1 - 2j/\ell|^r \leq 2^{1-\ell} \sum_{j=0}^{\lfloor \ell/2 \rfloor} \binom{\ell}{j} e^{-2rj/\ell} \leq 2^{1-\ell} (1 + e^{-2r/\ell})^{\ell}.$$

The first inequality pairs j with $\ell - j$ and uses $1 - x \leq e^{-x}$ on $0 \leq x \leq 1$. Multiply the regional bounds using independence. $\square$

6.2 Application to binary EA

The accumulator state must cross regional boundaries. Let

$$\mathbf{U}(X, Y) := \begin{pmatrix} 1 & XY \\ X & Y \end{pmatrix}. \quad (55)$$

For a uniform regional input of weight u , define

$$\mathbf{S}_{\ell, u}^{\text{EA}}(Y) := \begin{pmatrix} \ell \\ u \end{pmatrix}^{-1} [X^u] \mathbf{U}(X, Y)^{\ell}, \quad (56)$$

$$\mathbf{R}_{\ell, r}^{\text{EA}}(Y) := \sum_{u=0}^{\ell} \pi_{r, u}^{(\ell)} \mathbf{S}_{\ell, u}^{\text{EA}}(Y). \quad (57)$$

Theorem 6.4 (Exact left-regular EA first moment). *For $\mathbf{G} = \mathbf{B}\mathbf{A}$ with a region-stratified left-regular expander,*

$$\Pr_{\mathbf{B}}[\text{Bad}_L(\mathbf{G})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L [Y^h] \mathbf{e}_0^{\text{T}} \mathbf{R}_{\ell, r}^{\text{EA}}(Y)^{d_L} \mathbf{1}. \quad (58)$$

Proof. Lemma 6.1 expresses one regional input as a mixture of uniform Hamming shells. Matrix (56) averages the accumulator transfer over one shell. Matrix multiplication preserves the accumulator state between regions. Sum the low-weight coefficients and apply (12). $\square$

For intermediate supports, Lemma 6.2 bounds the output moment by

$$\gamma_r^{d_L} \mathbf{e}_0^\top \mathbf{T}_{a_r^{\text{Reg}}}(z)^n \mathbf{1}. \quad (59)$$

On a support interval $[u, v]$, bound the conditioning cost by $\gamma_v^{d_L}$. Spectral monotonicity from Section 4 bounds the remaining tail by $\sqrt{1+z} z^{-L} \lambda(a_u^{\text{Reg}}, z)^n$. Multiplying by a bound on $\sum_{r=u}^v \binom{k}{r}$ gives the intermediate interval contribution. For dense supports, invertibility of the accumulator and Lemma 6.3 give

$$\Pr[\text{wt}(\mathbf{xBA}) \leq L] \leq V_{2,n}(L) 2^{d_L - n} (1 + e^{-2r/\ell})^n, \quad V_{2,n}(L) := \sum_{h=0}^L \binom{n}{h}. \quad (60)$$

6.3 Application to binary EC

For a regional convolution input of weight u , define

$$\mathbf{S}_{\ell,u}^{\text{EC}}(Y) := \binom{\ell}{u}^{-1} [X^u] \mathbf{W}_m(X, Y)^\ell, \quad (61)$$

$$\mathbf{R}_{\ell,r}^{\text{EC}}(Y) := \sum_{u=0}^{\ell} \pi_{r,u}^{(\ell)} \mathbf{S}_{\ell,u}^{\text{EC}}(Y). \quad (62)$$

Theorem 6.5 (Exact left-regular wrapped-EC first moment). *Independently sample a region-stratified left-regular expander $\mathbf{B}$ and a wrapped convolution $\mathbf{C}$ of memory m . Then*

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L [Y^h] \mathbf{e}_m^\top \mathbf{R}_{\ell,r}^{\text{EC}}(Y)^{d_L} \mathbf{1}. \quad (63)$$

Proof. The proof is the regional version of Theorem 5.2. Lemma 6.1 gives the input mixture in each region. Matrix multiplication retains the convolution state at regional boundaries. The first-moment bound completes the proof. $\square$

The Poisson and dense bounds become

$$\mathbb{E}[z^{\text{wt}(\mathbf{xBC})}] \leq \gamma_r^{d_L} \mathbf{e}_m^\top \mathbf{N}_{m, a_r^{\text{Reg}}}(z)^n \mathbf{1}, \quad (64)$$

$$\Pr[\text{wt}(\mathbf{xBC}) \leq L] \leq V_{2,n}(L) 2^{d_L - n} (1 + e^{-2r/\ell})^n. \quad (65)$$

Equation (65) holds for every fixed invertible binary map $\mathbf{C}$ and therefore also after averaging the convolution. For an intermediate interval $[u, v]$, the verifier uses the factor $\gamma_v^{d_L}$ and the uniform transfer (42) with $a_- := a_u^{\text{Reg}}$ and $a_+ := a_v^{\text{Reg}}$. The dense bounds for EA and EC decrease with r , so their left-endpoint values bound each dense interval. In each case, multiply the tail bound by an upper bound on the number of messages in that interval.

6.4 Two-sided regular incidence over the binary field

Fix a right degree d_R and let $k = d_R \ell$. In each region, the two-sided regular expander partitions the k left vertices into ℓ groups of size d_R . A right coordinate is one when its group contains an odd number of vertices from the message support. Define

$$E_{d_R}(X) := \frac{(1+X)^{d_R} + (1-X)^{d_R}}{2}, \quad O_{d_R}(X) := \frac{(1+X)^{d_R} - (1-X)^{d_R}}{2}. \quad (66)$$

The coefficient of X^j in E_{d_R} is $\binom{d_R}{j}$ for even j and zero for odd j . The polynomial O_{d_R} gives the complementary odd coefficients.

Write the accumulator transfer as $\mathbf{U}(X, Y) = \mathbf{U}_0(Y) + X\mathbf{U}_1(Y)$. Similarly, write the wrapped-convolution transfer as $\mathbf{W}_m(X, Y) = \mathbf{W}_{m,0}(Y) + X\mathbf{W}_{m,1}(Y)$. For $R \in \{\text{EA}, \text{EC}\}$, let $(\mathbf{Q}_0, \mathbf{Q}_1)$ denote the corresponding pair of input-conditioned matrices. Define

$$\mathbf{R}_{\ell, d_R, r}^{\text{R}, \text{bi}}(Y) := \binom{k}{r}^{-1} [X^r] (E_{d_R}(X)\mathbf{Q}_0(Y) + O_{d_R}(X)\mathbf{Q}_1(Y))^\ell. \quad (67)$$

Theorem 6.6 (Exact two-sided regular binary first moment). *Let $n = d_L \ell$. Sample $\mathbf{B}$ from the two-sided regular ensemble by sampling the regional permutations independently. For binary EA,*

$$\Pr_{\mathbf{B}}[\text{Bad}_L(\mathbf{BA})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L [Y^h] \mathbf{e}_0^\top \mathbf{R}_{\ell, d_R, r}^{\text{EA}, \text{bi}}(Y)^{d_L} \mathbf{1}. \quad (68)$$

For binary wrapped EC, independently sample a wrapped convolution $\mathbf{C}$ of memory m and replace EA by EC in (67). Then

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L [Y^h] \mathbf{e}_m^\top \mathbf{R}_{\ell, d_R, r}^{\text{EC}, \text{bi}}(Y)^{d_L} \mathbf{1}. \quad (69)$$

Proof. Fix a message support of size r . Its positions in one regional permutation form a uniform r -subset of the k slots. Within one group, E_{d_R} counts selections that produce input zero, and O_{d_R} counts selections that produce input one. The coefficient of X^r in (67) counts all regional selections and retains the state of the recursive map. Division by $\binom{k}{r}$ averages over the uniform subset. The regional permutations are independent, so matrix multiplication composes their transfers. Summing over message supports and low output weights gives the two bounds. $\square$

The all-one message imposes a parity restriction that has no labeled large-field analogue.

Proposition 6.7 (Binary right-degree parity). *For a binary two-sided regular expander,*

$$\mathbf{1B} = (d_R \bmod 2)\mathbf{1}. \quad (70)$$

If d_R is even, both $\mathbf{BA}$ and $\mathbf{BC}$ are noninjective and have minimum distance zero.

Proof. Every right coordinate has exactly d_R incident edges. This proves (70). If d_R is even, the nonzero all-one message maps to zero before the invertible recursive map. $\square$

At rate one half, $d_L = 2d_R$. A usable binary two-sided regular ensemble therefore requires $d_L \equiv 2 \pmod{4}$. This excludes the current left-regular degrees 64 for EA and 28 for EC.

For right degree five, a local concentration argument removes the square-root loss from the positive coefficient saddle. Fix $r \in \{1, \dots, k-1\}$ and put $\theta_r := r/k$. Let

$$q_1(\theta) := \frac{1 - (1 - 2\theta)^5}{2}, \quad q_0(\theta) := 1 - q_1(\theta). \quad (71)$$

For $b \in \{0, 1\}$, define a random variable $Z_b \in \{0, 1, 2\}$ by

$$\Pr[Z_b = j] := \frac{\binom{5}{2j+b} \theta_r^{2j+b} (1 - \theta_r)^{5-2j-b}}{q_b(\theta_r)}. \quad (72)$$

Thus $2Z_b + b$ is the number of selected slots in one group, conditioned on parity b . Define

$$v_*(\theta_r) := \min_{b \in \{0,1\}} \text{Var}(Z_b), \quad \beta_{k,r} := \binom{k}{r} \theta_r^r (1 - \theta_r)^{k-r}, \quad (73)$$

$$M_{k,r} := \frac{\max\{q_0(\theta_r), q_1(\theta_r)\}^\ell}{\beta_{k,r}} \min \left\{ 1, \sqrt{\frac{\pi}{8\ell v_*(\theta_r)}} \right\}. \quad (74)$$

Lemma 6.8 (Point mass for right degree five). *Let $\mathbf{Y} \in \mathbb{F}_2^\ell$ be the image of a fixed weight- r support in one region of a two-sided regular expander with $d_R = 5$. Then*

$$\max_{\mathbf{y} \in \mathbb{F}_2^\ell} \Pr[\mathbf{Y} = \mathbf{y}] \leq M_{k,r}. \quad (75)$$

Proof. Independently select each of the k regional slots with probability θ_r . Conditioning on exactly r selected slots gives the uniform r -subset in the two-sided regular ensemble. Before conditioning, the group parities are independent and have probabilities $q_0(\theta_r)$ and $q_1(\theta_r)$.

Fix a parity vector $\mathbf{y}$ of weight u . Conditional on $\mathbf{Y} = \mathbf{y}$, the selected-slot count equals

$$u + 2 \sum_{i=1}^{\ell} Z_{y_i}. \quad (76)$$

The probability-generating polynomials of Z_0 and Z_1 , before normalization, are

$$\begin{aligned} (1 - \theta_r)^5 + 10\theta_r^2(1 - \theta_r)^3T + 5\theta_r^4(1 - \theta_r)T^2, \\ 5\theta_r(1 - \theta_r)^4 + 10\theta_r^3(1 - \theta_r)^2T + \theta_r^5T^2. \end{aligned}$$

Their discriminants are $80\theta_r^4(1 - \theta_r)^6$ and $80\theta_r^6(1 - \theta_r)^4$. Thus both polynomials have two real negative roots. After normalization, each Z_b has the distribution of a sum of two independent Bernoulli variables.

Consequently, the sum in (76) is a sum of independent Bernoulli variables with variance at least $\ell v_*(\theta_r)$. If such a sum has variance V , Fourier inversion gives

$$\max_j \Pr[Z = j] \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-2V \sin^2(t/2)} dt \leq \sqrt{\frac{\pi}{8V}}. \quad (77)$$

The last inequality uses $\sin(|t|/2) \geq |t|/\pi$ on this interval. Bayes' rule now gives

$$\Pr[\mathbf{Y} = \mathbf{y} \mid r \text{ selected slots}] \leq \frac{\max\{q_0(\theta_r), q_1(\theta_r)\}^\ell}{\beta_{k,r}} \min \left\{ 1, \sqrt{\frac{\pi}{8\ell v_*(\theta_r)}} \right\}.$$

This is (75). □

The certificate evaluates (75) in blocks. The following endpoint bound makes that evaluation rigorous. Put $r_{\text{mid}} := (k - 1)/2$ and $\theta_a := a/k$. For $k/5 \leq a \leq b \leq r_{\text{mid}}$, define

$$\widehat{M}_{a,b} := \frac{q_0(\theta_a)^\ell}{\beta_{k,b}} \sqrt{\frac{\pi}{8\ell \text{Var}(Z_1(\theta_a))}}. \quad (78)$$

Here $Z_1(\theta)$ denotes the variable from (72) with parameter θ .

Lemma 6.9 (Endpoint bound for the central range). *For $k/5 \leq a \leq b \leq r_{\text{mid}}$ and every fixed invertible $\mathbf{D}$,*

$$\sum_{r=a}^b \binom{k}{r} \Pr_{\mathbf{B}}[\text{wt}(\mathbf{x}_r \mathbf{B} \mathbf{D}) \leq L] \leq (b-a+1) \binom{k}{b} V_{2,n}(L) \widehat{M}_{a,b}^{d_L}, \quad (79)$$

where $\mathbf{x}_r$ is any fixed weight- r message. The same bound applies to the complementary supports $k-b \leq r \leq k-a$.

Proof. For $\theta \in [1/5, 1/2]$, $q_0(\theta)$ decreases. In the same interval, $v_*(\theta) = \text{Var}(Z_1(\theta))$, and this variance increases. To verify the latter claim, write

$$\text{Var}(Z_1(\theta)) = \frac{10\theta^2(1-\theta)^2(8\theta^4 - 24\theta^3 + 32\theta^2 - 20\theta + 5)}{(16\theta^4 - 40\theta^3 + 40\theta^2 - 20\theta + 5)^2}.$$

Its derivative has the same sign as

$$\theta(\theta-1)(2\theta-1)(4\theta^2-10\theta+5)(4\theta^4-20\theta^3+30\theta^2-20\theta+5),$$

which is nonnegative on $[1/5, 1/2]$. Indeed, after setting $t = 1 - 2\theta$, the last two factors become $t^2 + 3t + 1$ and $(t^4 + 6t^3 + 6t^2 + 6t + 1)/4$. The same substitution gives

$$\text{Var}(Z_0(\theta)) - \text{Var}(Z_1(\theta)) = \frac{5t^3(1-t)^2(1+t)^2(t^2+1)(2t^4+t^2+2)}{4(t^4-t^3+t^2-t+1)^2(t^4+t^3+t^2+t+1)^2},$$

which is nonnegative for $t \in [0, 3/5]$. Hence $v_*(\theta) = \text{Var}(Z_1(\theta))$ throughout the interval.

It remains to control $\beta_{k,r}$. Its consecutive ratio is

$$\frac{\beta_{k,r+1}}{\beta_{k,r}} = \left(1 + \frac{1}{r}\right)^r \left(1 - \frac{1}{k-r}\right)^{k-r-1} \leq 1 \quad (r \leq r_{\text{mid}}).$$

Here the second factor is the reciprocal of $(1 + 1/(k-r-1))^{k-r-1}$. The inequality follows because $(1 + 1/s)^s$ increases with s and $r \leq k-r-1$. Consequently, every point mass in a block is at most $\widehat{M}_{a,b}^{d_L}$. The message count is at most $\binom{k}{b}$. Lemma 6.8, invertibility of $\mathbf{D}$, and a sum over the block prove (79). Replacing a selected set by its complement exchanges even and odd group parities. All terms in the bound are therefore unchanged. $\square$

Corollary 6.10 (Dense degree-five bound). *Let $\mathbf{D}$ be any fixed invertible binary map on $\mathbb{F}_2^n$. For a fixed weight- r message $\mathbf{x}$,*

$$\Pr_{\mathbf{B}}[\text{wt}(\mathbf{x} \mathbf{B} \mathbf{D}) \leq L] \leq V_{2,n}(L) M_{k,r}^{d_L}. \quad (80)$$

The same bound holds when $\mathbf{D}$ is sampled independently of $\mathbf{B}$.

Proof. The regional permutations are independent, so every point probability of $\mathbf{x} \mathbf{B}$ is at most $M_{k,r}^{d_L}$. Invertibility gives exactly $V_{2,n}(L)$ expanded vectors whose image under $\mathbf{D}$ has weight at most L . Summing their point probabilities proves the fixed-map claim. Averaging over an independent $\mathbf{D}$ proves the second claim. $\square$

Right degree three admits the same local argument with a simpler conditional distribution. For $\theta := r/k$, define

$$q_1^{(3)}(\theta) := \frac{1 - (1 - 2\theta)^3}{2}, \quad q_0^{(3)}(\theta) := 1 - q_1^{(3)}(\theta). \quad (81)$$

For $s \in \{0, 1\}$, let $Z_s^{(3)}(\theta) \in \{0, 1\}$ satisfy

$$\Pr[Z_s^{(3)}(\theta) = j] := \frac{\binom{3}{2j+s} \theta^{2j+s} (1-\theta)^{3-2j-s}}{q_s^{(3)}(\theta)}. \quad (82)$$

Thus $2Z_s^{(3)}(\theta) + s$ is the selected-slot count in a three-slot group, conditioned on its parity. Put

$$M_{k,r}^{(3)} := \frac{\max\{q_0^{(3)}(\theta), q_1^{(3)}(\theta)\}^\ell}{\beta_{k,r}} \min \left\{ 1, \sqrt{\frac{\pi}{8\ell \min_s \text{Var}(Z_s^{(3)}(\theta))}} \right\}. \quad (83)$$

Lemma 6.11 (Central blocks for right degree three). *Let $r_{\text{mid}} = (k-1)/2$, and let $1 \leq a \leq b \leq r_{\text{mid}}$. Suppose*

$$\sqrt{\frac{\pi}{8\ell \text{Var}(Z_1^{(3)}(a/k))}} \leq 1.$$

Define

$$\widehat{M}_{a,b}^{(3)} := \frac{q_0^{(3)}(a/k)^\ell}{\beta_{k,b}} \sqrt{\frac{\pi}{8\ell \text{Var}(Z_1^{(3)}(a/k))}}. \quad (84)$$

Then, for every fixed invertible $\mathbf{D}$,

$$\sum_{r=a}^b \binom{k}{r} \Pr_{\mathbf{B}}[\text{wt}(\mathbf{x}_r \mathbf{B} \mathbf{D}) \leq L] \leq (b-a+1) \binom{k}{b} V_{2,n}(L) (\widehat{M}_{a,b}^{(3)})^{d_L}. \quad (85)$$

The same bound applies to supports $k-b \leq r \leq k-a$.

Proof. Conditioning independent slot selections on their total count gives a uniform weight- r support. Conditional on the parity vector, the remaining count is a sum of the independent Bernoulli variables in (82). The Fourier bound (77) therefore gives

$$\max_{\mathbf{y} \in \mathbb{F}_2^\ell} \Pr[\mathbf{Y} = \mathbf{y}] \leq M_{k,r}^{(3)}.$$

For $0 < \theta \leq 1/2$, direct calculation gives

$$\begin{aligned} \text{Var}(Z_0^{(3)}(\theta)) &= \frac{3\theta^2(1-\theta)^2}{(4\theta^2 - 2\theta + 1)^2}, \\ \text{Var}(Z_1^{(3)}(\theta)) &= \frac{3\theta^2(1-\theta)^2}{(4\theta^2 - 6\theta + 3)^2}. \end{aligned}$$

The second variance is no larger than the first and increases on this interval. Indeed, their difference has the sign of $1 - 2\theta$, and

$$\frac{d}{d\theta} \text{Var}(Z_1^{(3)}) = \frac{-6\theta(\theta-1)(2\theta^2 - 6\theta + 3)}{(4\theta^2 - 6\theta + 3)^3} \geq 0.$$

Also, $q_0^{(3)}(\theta)$ decreases for $\theta \leq 1/2$, and $\beta_{k,r}$ decreases for integer $r \leq r_{\text{mid}}$, as shown in the proof of Lemma 6.9. Hence every regional point mass in the block is at most $\widehat{M}_{a,b}^{(3)}$. Independence of the regions, invertibility of $\mathbf{D}$, and a sum over the block prove (85). Complementing all selected slots exchanges the two group parities and proves the final claim. $\square$

The same argument extends to every odd right degree. The following form is weaker than the specialized degree-three and degree-five bounds, but it is sufficient for the intermediate finite profiles. Let $d = 2h + 1 \geq 3$ and define

$$\begin{aligned}\Theta_0(d) &:= \left\{ \frac{(2j+1)\pi}{2d} : 0 \leq j < h \right\}, & \Theta_1(d) &:= \left\{ \frac{j\pi}{d} : 1 \leq j \leq h \right\}, \\ f(t) &:= \frac{t}{(1+t)^2}, & c_a &:= \frac{k-a}{a}.\end{aligned}$$

For $s \in \{0, 1\}$, put

$$v_{d,s}(a) := \sum_{\theta \in \Theta_s(d)} \min \{ f(c_a^2 \tan^2 \theta), f(\tan^2 \theta) \}, \quad v_d(a) := \min_s v_{d,s}(a). \quad (86)$$

Lemma 6.12 (Central blocks for odd right degree). *Let $d_R = d \geq 3$ be odd, let $r_{\text{mid}} = (k-1)/2$, and let $1 \leq a \leq b \leq r_{\text{mid}}$. Define $\theta_a := a/k$ and*

$$\widehat{M}_{a,b}^{(d)} := \frac{q_{0,d}(\theta_a)^\ell}{\beta_{k,b}} \sqrt{\frac{\pi}{8\ell v_d(a)}}, \quad q_{0,d}(\theta) := \frac{1 + (1-2\theta)^d}{2}. \quad (87)$$

Suppose the square-root factor in (87) is at most one. For every fixed invertible $\mathbf{D}$,

$$\sum_{r=a}^b \binom{k}{r} \Pr_{\mathbf{B}}[\text{wt}(\mathbf{x}_r \mathbf{B} \mathbf{D}) \leq L] \leq (b-a+1) \binom{k}{b} V_{2,n}(L) (\widehat{M}_{a,b}^{(d)})^{d_L}. \quad (88)$$

The same bound applies to supports $k-b \leq r \leq k-a$.

Proof. Fix $\theta \in [\theta_a, 1/2]$. In one right group, let J count independently selected slots. Conditional on $J \bmod 2 = s$, the probability-generating polynomial of $(J-s)/2$ is proportional to

$$\sum_{j=0}^h \binom{d}{2j+s} \theta^{2j+s} (1-\theta)^{d-2j-s} T^j.$$

For $s = 0$, its roots are

$$-\left(\frac{1-\theta}{\theta}\right)^2 \tan^2 \varphi, \quad \varphi \in \Theta_0(d).$$

For $s = 1$, the same formula holds with $\varphi \in \Theta_1(d)$. These identities follow by taking the even part of $(1-\theta+\theta\sqrt{T})^d$ and by dividing its odd part by $\sqrt{T}$.

All roots are negative. After normalization, each polynomial is therefore the generating polynomial of a sum of independent Bernoulli variables. A root $-\rho$ contributes variance $f(\rho)$. As θ runs from θ_a to $1/2$, each ρ changes monotonically between $c_a^2 \tan^2 \varphi$ and $\tan^2 \varphi$. The function f increases on $(0, 1]$ and decreases on $[1, \infty)$. Its minimum on this interval is at an endpoint. Hence either conditional half-count has variance at least $v_d(a)$.

The Fourier estimate (77) now bounds the conditional total-count point mass. For $\theta \leq 1/2$, $q_{0,d}(\theta)$ is the larger parity probability and decreases with θ . Also, $\beta_{k,r}$ decreases for $r \leq r_{\text{mid}}$. The endpoint argument from Lemma 6.9 proves (88). Complementing all slots exchanges the two parity laws and proves the final claim. $\square$

6.5 Certified improvements at rate one half

Theorem 6.13 (Regular binary EA and EC certificates). *The following claims hold.*

1. Let $k = 1,048,576$, $d_L = 64$, $\ell = 32,768$, $n = d_L \ell = 2k$, and $L = 230,718$. Sample $\mathbf{B}$ from the region-stratified left-regular ensemble. Then

$$\Pr_{\mathbf{B}}[\text{Bad}_L(\mathbf{BA})] \leq 8.982345581 \cdot 10^{-7} < 2^{-20.0864}. \quad (89)$$

2. Let $k = 1,048,572$, $d_L = 28$, $\ell = 74,898$, $n = d_L \ell = 2k$, and $L = 230,730$. Sample $\mathbf{B}$ from the region-stratified left-regular ensemble and independently sample a wrapped convolution $\mathbf{C}$ of memory 9. Then

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] < 2.036627 \cdot 10^{-7} < 2^{-22.2273}. \quad (90)$$

3. Let $k = 1,048,575$, $d_L = 10$, $d_R = 5$, $\ell = 209,715$, $n = 2k$, and $L = 230,729$. Sample $\mathbf{B}$ from the two-sided regular ensemble and independently sample a wrapped convolution $\mathbf{C}$ of memory 15. Then

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] < 1.6475000 \cdot 10^{-10} < 2^{-32.4990}. \quad (91)$$

4. Let $k = 1,048,635$, $d_L = 14$, $d_R = 7$, $\ell = 149,805$, $n = 2k$, and $L = 230,742$. Sample $\mathbf{B}$ from the two-sided regular ensemble and independently sample a wrapped convolution $\mathbf{C}$ of memory 6. Then

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] < 1.402384 \cdot 10^{-8} < 2^{-26.0875}. \quad (92)$$

5. Let $k = 1,048,635$, $d_L = 18$, $d_R = 9$, $\ell = 116,515$, $n = 2k$, and $L = 230,742$. Sample $\mathbf{B}$ from the two-sided regular ensemble and independently sample a wrapped convolution $\mathbf{C}$ of memory 4. Then

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] < 2.987318 \cdot 10^{-7} < 2^{-21.6746}. \quad (93)$$

6. Let $k = 1,048,575$, $d_L = 6$, $d_R = 3$, $\ell = 349,525$, $n = 2k$, and $L = 230,729$. Sample $\mathbf{B}$ from the two-sided regular ensemble and independently sample a wrapped convolution $\mathbf{C}$ of memory 79. Then

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] < 3.558444 \cdot 10^{-7} < 2^{-21.4222}. \quad (94)$$

Proof. For EA, the verifier combines the exact regional transfer for small supports, (59) for intermediate supports, and (60) for dense supports. For EC it applies the corresponding three bounds from (63)–(65). The support intervals cover $[1, k]$.

For the degree-10/5 EC result, the verifier uses exact regional transfers for supports 1 through 16. It uses 111 positive coefficient blocks for the two outer ranges and 47 applications of Lemma 6.9 for the central range. It checks the all-one support directly. Outward-rounded Arb arithmetic gives the displayed values. Section 8 gives the verifier details.

For the degree-14/7 and degree-18/9 results, exact regional transfers cover supports 1 through 24 and 1 through 32, respectively. Positive coefficient blocks cover the two outer ranges. The central ranges use Lemma 6.12. A separate transfer checks each all-one support.

For the degree-6/3 result, exact regional transfers cover supports 1 through 383. A layered representation allows each reachable matrix entry to use an independent power-of-two scale. The representation uses only positive operations and rounds every operation upward. It therefore retains entries below the ordinary binary64 range without adding a matrix-wide absolute error. The verifier

Table 5: Certified binary recursive-map frontier at rate one half.

map	expander	d_L	d_R	memory m	$d_L + m$	failure bits
EA	left regular	64	–	1	65	20.0864
EC	two-sided regular	18	9	4	22	21.6746
EC	two-sided regular	14	7	6	20	26.0875
EC	two-sided regular	10	5	15	25	32.4990
EC	two-sided regular	6	3	79	85	21.4222

uses 103 coefficient blocks outside the central range and 28 applications of Lemma 6.11 inside that range. It checks the all-one support separately. Section 8 gives the resulting bounds. $\square$

The EA cutoff is 0.9998823119 times the binary rate-one-half GV distance. A matched Bernoulli certificate requires expected degree 75. Left regularity therefore removes 14.67% of the expander edges.

The EC cutoff is 0.9999381317 times the binary rate-one-half GV distance. The comparable Bernoulli memory-9 profile has expected degree 42.2127. Replacing it by degree 28 removes 33.67% of the expander edges while losing eleven units of output weight.

The two-sided EC cutoff is 0.9999309371 times the binary rate-one-half GV distance. Relative to the left-regular degree-28 profile, two-sided regularity and six additional memory words remove 64.29% of the expander edges. The simple work proxy $d_L + m$ decreases from 37 to 25.

Table 5 gives the certified binary frontier. We record the accumulator as a memory-one recursive map because it retains only y_{t-1} . Over $\mathbb{F}_2$, this map is also the memory-one instance of the wrapped convolution in (8). Its certified expander is left regular; the four other rows use two-sided regular incidence. A dash in the d_R column means that the distribution does not fix the right degree. The degree-14/7, memory-6 profile minimizes the proxy $d_L + m$, with value 20. Reducing the left degree to 10 or 6 requires substantially more convolution memory.

At rate one half, the degrees satisfy $d_L = 2d_R$. The binary construction also requires odd d_R : if d_R is even, the all-one message contributes an even number of ones to every right vertex and therefore encodes to zero. Consequently, degree 8/4 is not an intermediate candidate between 6/3 and 10/5. The usable left degrees at rate one half are congruent to two modulo four; Table 5 lists four such EC profiles.

The EA outcome is different. At the same cutoff, the $d_L/d_R = 62/31$ diagnostic is dominated by message weight two and does not reach a 2^{-20} first moment. Fixing the right degrees does not recover the two removed EA edges.

The degree-6/3 calculation also illustrates a numerical issue in exact enumerator proofs. Some reachable transfer entries lie below the ordinary binary64 range. A global underflow allowance becomes useless after the union bound multiplies it by the number of messages. The layered verifier avoids this loss by retaining each exponent band as a separate positive matrix. The matrix products use relative upward rounding; Arb evaluates the remaining scalar expressions.

6.6 Scalar extension preserves distance but not field mixing

The binary generators above can act on messages over an extension field of $\mathbb{F}_2$, and this reinterpretation preserves minimum distance. It does not, however, turn a binary encoder into a field-mixing encoder.

Lemma 6.14 (Distance under scalar extension). *Let $\mathbf{G} \in \mathbb{F}_2^{k \times n}$, and let K be a finite extension of $\mathbb{F}_2$. View the entries of $\mathbf{G}$ as elements of K . Then*

$$\min_{\mathbf{x} \in K^k \setminus \{0\}} \text{wt}(\mathbf{x}\mathbf{G}) = \min_{\mathbf{x} \in \mathbb{F}_2^k \setminus \{0\}} \text{wt}(\mathbf{x}\mathbf{G}). \quad (95)$$

Proof. Fix an $\mathbb{F}_2$ -basis $\beta_1, \dots, \beta_t$ of K . Every $\mathbf{x} \in K^k$ has a unique decomposition

$$\mathbf{x} = \sum_{j=1}^t \beta_j \mathbf{x}_j, \quad \mathbf{x}_j \in \mathbb{F}_2^k.$$

Because $\mathbf{G}$ is binary, $\mathbf{x}\mathbf{G} = \sum_j \beta_j (\mathbf{x}_j \mathbf{G})$. A coordinate of $\mathbf{x}\mathbf{G}$ is zero exactly when that coordinate is zero in every $\mathbf{x}_j \mathbf{G}$. Hence

$$\text{supp}(\mathbf{x}\mathbf{G}) = \bigcup_{j=1}^t \text{supp}(\mathbf{x}_j \mathbf{G}).$$

If $\mathbf{x} \neq 0$, some $\mathbf{x}_j$ is nonzero. The union therefore has size at least the binary minimum distance. Conversely, binary messages form a subset of K^k , which gives the reverse inequality. $\square$

Indeed, the proof writes the extension-field message as t binary component messages and applies the same binary matrix to each component independently. This decomposition is useful for binary-choice Silent OT. It is not the mixing required by a field-valued Silent VOLE instance. For that application we instead use the labeled finite-field construction of Section 7, whose edge labels and convolution coefficients are sampled in the ambient field.

7 Finite fields and field-size scaling

Over $\mathbb{F}_2$, a nonzero message is determined by its support. Over $\mathbb{F}_p$, two messages with the same support can cancel differently. An addition-only expander therefore has no support-only output law. We restore that symmetry by assigning an independent label from $\mathbb{F}_p^*$ to every edge.

Fix a prime power p , choose the field $\mathbb{F}_p$, and write $s := p - 1$. Multiplying a fixed nonzero message value by a uniform edge label gives a uniform element of $\mathbb{F}_p^*$. Thus random labels erase the message values while preserving its support.

The analysis has two scales. The asymptotic theorem below fixes p while the code length grows. The finite certificates in Section 8 instead fix a cryptographic-size length and fields whose orders are near 2^{128} . In that finite regime, the proof must control the projective message count rather than absorb it into an asymptotic exponent.

7.1 Labeled Bernoulli incidence

In the labeled Bernoulli ensemble, an entry is zero with probability $1 - \rho$ and equals each element of $\mathbb{F}_p^*$ with probability ρ/s .

Lemma 7.1 (Labeled Bernoulli output law). *Fix $\mathbf{x} \in \mathbb{F}_p^k$ with support size r . The coordinates of $\mathbf{x}\mathbf{B}$ are independent. Each coordinate is nonzero with probability*

$$a_r^{(p)} := \frac{s}{p} \left(1 - \left(1 - \frac{p\rho}{s} \right)^r \right). \quad (96)$$

Conditional on being nonzero, its value is uniform in $\mathbb{F}_p^$.*

Proof. Let χ be a nonconstant additive character of $\mathbb{F}_p$, that is, a complex-valued function satisfying $\chi(x+y) = \chi(x)\chi(y)$. One active row contribution Z satisfies

$$\mathbb{E}[\chi(Z)] = 1 - \rho + \frac{\rho}{s} \sum_{a \in \mathbb{F}_p^*} \chi(a) = 1 - \frac{p\rho}{s}.$$

The r active rows are independent. Fourier inversion over the additive group of $\mathbb{F}_p$ gives (96) and equal mass at every nonzero value. Different columns use independent samples. $\square$

7.2 Finite-field EA

Suppose an accumulator input is zero with probability $1 - a$ and is otherwise uniform in $\mathbb{F}_p^*$. Group the accumulator states into zero and nonzero. These two classes have the exact transfer matrix

$$\mathbf{T}_a^{(p)}(Y) := \begin{pmatrix} 1 - a & aY \\ a/s & (1 - a/s)Y \end{pmatrix}. \quad (97)$$

From a nonzero state, exactly one of the s possible nonzero increments returns to zero. Hence this two-state reduction is exact.

All nonzero scalar multiples of one message produce codewords with the same support. A *projective message line* is a one-dimensional subspace of $\mathbb{F}_p^k$. The number of such lines with support size r is

$$N_{p,k,r} := \binom{k}{r} s^{r-1}. \quad (98)$$

Theorem 7.2 (Labeled Bernoulli EA first moment). *Sample a labeled Bernoulli expander $\mathbf{B}$ and set $\mathbf{G} := \mathbf{BA}$. Then*

$$\Pr_{\mathbf{B}}[\text{Bad}_L(\mathbf{G})] \leq \sum_{r=1}^k N_{p,k,r} \sum_{h=0}^L [Y^h] \mathbf{e}_0^T \mathbf{T}_{a_r}^{(p)}(Y)^n \mathbf{1}. \quad (99)$$

Proof. Lemma 7.1 gives the exact two-class input law for every projective line with support size r . Matrix (97) gives its output-weight generating function. Sum over projective lines and apply the first-moment bound. Counting one representative per line is sufficient because scalar multiplication does not change output support. $\square$

For $a \in [0, s/p]$ and $z \in (0, 1]$, define

$$\lambda_p(a, z) := \frac{1}{2} \left(1 - a + (1 - a/s)z + \sqrt{(1 - a - (1 - a/s)z)^2 + 4a^2z/s} \right), \quad (100)$$

$$J_\delta^{(p)}(a) := \sup_{0 < z \leq 1} \{ \delta \ln z - \ln \lambda_p(a, z) \}, \quad (101)$$

$$I_{p,\delta} := \left(\sqrt{1 - \delta} - \sqrt{\frac{\delta}{p - 1}} \right)^2. \quad (102)$$

Lemma 7.3 (Finite-field accumulator endpoints). *For $0 < \delta < s/p$,*

$$\Pr[H \leq \lfloor \delta n \rfloor] \leq \sqrt{p} \exp(-n J_\delta^{(p)}(a)), \quad (103)$$

$$\liminf_{a \downarrow 0} \frac{J_\delta^{(p)}(a)}{a} \geq I_{p,\delta}, \quad (104)$$

$$J_\delta^{(p)}(s/p) = \ln p - h_p(\delta). \quad (105)$$

Here H is the output weight for independent inputs that are nonzero with probability a and uniform conditional on being nonzero. The function $J_\delta^{(p)}$ is continuous and positive on $(0, s/p]$.

Proof. Conjugating (97) by $\text{diag}(1, \sqrt{sz})$ gives a symmetric positive semidefinite matrix with largest eigenvalue $\lambda_p(a, z)$. The same positive-marker argument as in Lemma 4.2 gives (103).

For the sparse endpoint, set $z = e^{-\theta a}$ and expand at $a = 0$. Optimizing the coefficient of a gives $(\sqrt{1-\delta} - \sqrt{\delta/s})^2$. At $a = s/p$, the input and accumulator state are uniform in $\mathbb{F}_p$, and $\lambda_p(s/p, z) = (1 + sz)/p$. The optimizer $z = \delta/(s(1-\delta))$ gives (105).

For $a > 0$, the stationary nonzero-state probability is s/p . Consequently, the derivative of $\ln \lambda_p(a, z)$ with respect to $\ln z$ at $z = 1$ equals $s/p > \delta$. A marker slightly below one therefore gives a positive exponent. Also, $\lambda_p(a, z) \geq 1 - a \geq 1/p$, so the defining objective is at most $\delta \ln z + \ln p$. This tends uniformly to $-\infty$ as $z \downarrow 0$. Restricting to a compact interval of positive markers proves continuity, and hence a uniformly positive exponent on each compact subinterval of $(0, s/p]$. $\square$

Theorem 7.4 (Fixed-field EA distance). *Fix a prime power p and the field $\mathbb{F}_p$. Let $k/n \rightarrow R \in (0, 1)$ and sample a labeled Bernoulli expander of density $\rho_n = C \ln(n)/n$. If $0 < \delta < (p-1)/p$ and*

$$CI_{p,\delta} > 1, \quad (106)$$

$$h_p(\delta) < (1-R) \ln p, \quad (107)$$

then

$$\Pr_{\mathbf{B}}[d_{\min}(\mathbf{BA}) \leq \lfloor \delta n \rfloor] = o(1). \quad (108)$$

Proof. Use the fixed thresholds from the proof of Theorem 4.3, replacing I_δ by $I_{p,\delta}$. For sufficiently large n ,

$$r\rho_n(1 - r\rho_n) \leq \frac{s}{p}(1 - e^{-pr\rho_n/s}) \leq a_r^{(p)} \leq r\rho_n,$$

where the first inequality uses $p/s \leq 2$. Thus the weight- r tail is at most $\sqrt{p}n^{-\gamma r}$ throughout $r\rho_n \leq \tau$, for some fixed $\gamma > 1$. Since p is fixed and $N_{p,k,r} \leq (sk)^r$, the sparse sum vanishes. For $r\rho_n > \tau$, the activation probability is at least $(s/p)(1 - e^{-p\tau/s}) > 0$. For intermediate supports, the rate function is uniformly positive and dominates their small counting exponent. For $r > \eta n$, one has $a_r^{(p)} = s/p - o(1)$ uniformly. There are at most p^k/s projective lines, so their total exponent is

$$R \ln p + h_p(\delta) - \ln p + o(1),$$

which is negative by (107). $\square$

7.3 Reaching GV with tunable sampling failure

The preceding distance theorems fix a gap below GV. We now let the gap shrink while retaining logarithmic expected degree. Increasing its constant gives any fixed polynomial failure exponent. A further logarithmic factor in the degree makes failure negligible in n , meaning smaller than n^{-A} for every fixed $A > 0$ for all sufficiently large n .

Theorem 7.5 (GV distance with logarithmic expected degree). *Fix a finite field $\mathbb{F}_p$ and a limiting rate $R \in (0, 1)$. Use either the accumulator $\mathbf{T} := \mathbf{A}$ or, when $p = 2$, a sampled wrapped convolution $\mathbf{T} := \mathbf{C}$ with fixed memory $m \geq 1$. For every $A > 0$, there is a constant $C > 0$, depending only on p, R, A and the chosen recursive map's memory, with the following property. Let $R_n := k/n \rightarrow R$,*

sample a labeled Bernoulli expander of density $\rho_n := C \ln(n)/n$ independently of $\mathbf{T}$, and set $\mathbf{G} := \mathbf{B}\mathbf{T}$. For all sufficiently large n , define

$$L_n := \left\lfloor n \left(\delta_{p,\text{GV}}(R_n) - \frac{1}{\ln n} \right) \right\rfloor. \quad (109)$$

Then, for all sufficiently large n ,

$$\Pr_{\mathbf{G}}[\text{Bad}_{L_n}(\mathbf{G})] \leq n^{-A}. \quad (110)$$

Encoding with the stored sparse expander uses $O(n \log n)$ expected field operations. The length threshold may depend on the sequence R_n .

Proof. Write $s := p - 1$. Choose a fixed $\bar{\delta}$ strictly between $\delta_{p,\text{GV}}(R)$ and s/p . For all sufficiently large n , $L_n/n < \bar{\delta}$. For EA, Lemma 7.3 gives a tail $K \exp(-bna)$ at cutoff $\bar{\delta}n$ for sufficiently small activation a , where $K, b > 0$ are constants. For binary wrapped EC, use (161) instead. Both results also give a uniform exponential tail when activation is bounded away from zero.

Choose a fixed $\tau \in (0, 1)$ small enough for the sparse tail bound. For $r\rho_n \leq \tau$, the activation inequalities used in Theorem 7.4 give $a_r^{(p)} \geq (1 - \tau)r\rho_n$. Thus the tail for a fixed weight- r message is at most $Kn^{-b'C_r}$, where $b' := b(1 - \tau) > 0$ does not depend on C . There are at most $(sk)^r$ projective message lines of support size r . Choose C so that $b'C > A + 1$. Since $k \leq n$ for all sufficiently large n , the sparse contribution is at most

$$K \sum_{r \geq 1} (sn^{1-b'C})^r = O(n^{1-b'C}) = o(n^{-A}).$$

For $r\rho_n > \tau$, activation is at least $(s/p)(1 - e^{-p\tau/s}) > 0$. Choose a fixed $\eta \in (0, R/2)$ small enough that the number of messages with $r \leq \eta n$ has a smaller exponential rate than the uniform tail bound. This is possible because its exponent per output coordinate is at most $Rh(\eta/R) + \eta \ln s + o(1)$, which tends to zero with η . The intermediate contribution is therefore $\exp(-\Omega(n))$.

For $r > \eta n$, put $t_n := 1 - p\rho_n/s$; for sufficiently large n , $0 < t_n < 1$. Each expander-output coordinate has maximum point probability $(1 + st_n^r)/p$, attained at zero. The coordinates are independent and $t_n^r \leq n^{-\beta}$, where $\beta := (p/s)\eta C > 0$. Condition on any sampled recursive map. It is invertible and independent of $\mathbf{B}$, so a Hamming ball has exactly its cardinality many preimages. The positive-marker bound on $(1 + sz)^n$ gives ball volume at most $\exp(nh_p(L_n/n))$. Union-bounding at most p^k messages therefore bounds the dense contribution by

$$\exp\left(n \left[h_p(L_n/n) - (1 - R_n) \ln p + \ln(1 + sn^{-\beta}) \right]\right).$$

The derivative $h'_p(x) = \ln(s(1 - x)/x)$ is uniformly positive near $\delta_{p,\text{GV}}(R_n)$, since $R_n \rightarrow R \in (0, 1)$. The entropy gap in this display is consequently at most $-b''/\ln n$ for a constant $b'' > 0$. The correction from t_n^r is $o(1/\ln n)$, so the dense contribution is $\exp(-\Omega(n/\ln n))$. The dense bound holds for every recursive map and hence after averaging over its randomness. The three contributions sum to $o(n^{-A})$, which proves (110) for all sufficiently large n . The expected number of expander edges is $kC \ln n$; the recursive map uses $O(n)$ field operations because its memory is fixed. This proves the cost claim. $\square$

Outside the bad event, the relative distance exceeds the GV value minus $1/\ln n + 1/n$. Thus logarithmic expected degree already attains the full asymptotic GV benchmark. The proof permits C proportional to $A + 1$, with the proportionality constant depending on the field, rate, and recursive map. It does not give one fixed C that works for all failure exponents A .

Corollary 7.6 (GV distance with negligible sampling failure). *Use the field, rate sequence, recursive map, and cutoff (109) from Theorem 7.5. Instead sample the independent labeled Bernoulli expander with density $\rho_n := (\ln n)^2/n$, and set $\mathbf{G} := \mathbf{BT}$. There is a constant $c > 0$ such that, for all sufficiently large n ,*

$$\Pr_{\mathbf{G}}[\text{Bad}_{L_n}(\mathbf{G})] \leq \exp(-c(\ln n)^2). \quad (111)$$

Encoding with the stored sparse expander uses $O(n \log^2 n)$ expected field operations.

Proof. Use the fixed $\bar{\delta}, \tau, \eta, K, b'$ from the preceding proof. For sparse supports, the per-message tail is now $K \exp(-b'r(\ln n)^2)$. Multiplying by $(sk)^r$ and summing gives $\exp(-\Omega((\ln n)^2))$. The intermediate contribution remains $\exp(-\Omega(n))$. For dense supports, the point-mass correction is at most

$$\ln\left(1 + se^{-(p/s)\eta(\ln n)^2}\right) = o(1/\ln n).$$

The same entropy gap therefore gives $\exp(-\Omega(n/\ln n))$ for this contribution. Summing proves (111). The expected edge count is $k(\ln n)^2$, which gives the stated cost. $\square$

The extra logarithmic factor strengthens the failure bound, not the distance target. Neither result asserts a guarantee at the exact finite-length GV cutoff. The two failure guarantees are genuinely different for Bernoulli incidence: at density $C \ln(n)/n$ with fixed C , a specified row is zero with probability $(1 - \rho_n)^n = n^{-C}(1 + o(1))$. That event makes the generator noninjective, and its probability is not negligible in n .

The phrase “fix a field” is essential. If p grows with n , the factor s^{r-1} in (98) changes the sparse counting exponent. Theorem 7.4 therefore does not justify a cryptographic-size field at finite length. The next bounds group projective messages more carefully.

7.4 Constraint traces for the projective union

For a fixed support S , the elementary union bound multiplies a per-line failure probability by $s^{|S|-1}$. That multiplier can exceed one long before the underlying event becomes likely. The first correction is to cap the union for each support:

$$\Pr[\text{Bad}_L(\mathbf{BA})] \leq \sum_{r=1}^k \binom{k}{r} \min\{1, s^{r-1} \tau_r(L)\}, \quad (112)$$

where $\tau_r(L)$ is the bad-output probability of one projective line. Here RAA denotes repeat–accumulate–accumulate codes. This support grouping is related to recent large-field RAA analyses [AZ26, Kha26].

A sharper bound counts the independent field equations imposed by one output trace. Call an intermediate coordinate *occupied* when at least one edge from the message support reaches it. In the Bernoulli ensemble, an occupied indicator has probability

$$u_r := 1 - (1 - \rho)^r. \quad (113)$$

Let z mark a nonzero accumulator output and let v mark an occupied coordinate whose new accumulator state is zero. Define

$$\mathbf{K}_u(z, v) := \begin{pmatrix} 1 - u + uv & uz \\ uv & z \end{pmatrix}. \quad (114)$$

Theorem 7.7 (Bernoulli constraint-trace bound). *For the labeled Bernoulli EA ensemble,*

$$\Pr[\text{Bad}_L(\mathbf{BA})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L \sum_{a=0}^n [z^h v^a] \mathbf{e}_0^\top \mathbf{K}_{u_r}(z, v)^n \mathbf{1} \min\{1, s^{r-1-a}\}. \quad (115)$$

Proof. Fix a message support S of size r , the occupied coordinates, and one zero–nonzero accumulator trace. Suppose a occupied coordinates end in state zero. Each such event gives one prefix equation in the active edge labels. Subtracting consecutive equations isolates a nonempty set of fresh labels, so the a equations have rank a . One projective line satisfies them with probability at most s^{-a} . Union-bound the s^{r-1} lines on S , cap the result at one, and average the traces with (114). $\square$

The factor $\min\{1, s^{r-1-a}\}$ is the key cancellation. A trace with more independent equations than projective message degrees of freedom gains a field factor for every surplus equation. A trace with too few equations is charged at most one, not the full projective count.

7.5 Two-sided regular trace for EA

For the two-sided regular ensemble, each right coordinate has d_R incident slots. A support of size r maps to a uniform r -subset of the $k = d_R \ell$ regional slots. Define

$$\mathbf{Q}_0(z) := \begin{pmatrix} 1 & 0 \\ 0 & z \end{pmatrix}, \quad \mathbf{Q}_1(z, v) := \begin{pmatrix} v & z \\ v & z \end{pmatrix}, \quad (116)$$

and

$$\mathbf{R}_{\ell, d_R, r}^{\text{EA}}(z, v) := \binom{k}{r}^{-1} [X^r] \left(\mathbf{Q}_0(z) + ((1 + X)^{d_R} - 1) \mathbf{Q}_1(z, v) \right)^\ell. \quad (117)$$

Theorem 7.8 (Two-sided regular EA trace). *For the labeled two-sided regular EA ensemble,*

$$\Pr[\text{Bad}_L(\mathbf{BA})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L \sum_{a=0}^n [z^h v^a] \mathbf{e}_0^\top \mathbf{R}_{\ell, d_R, r}^{\text{EA}}(z, v)^{d_L} \mathbf{1} \min\{1, s^{r-1-a}\}. \quad (118)$$

Proof. In one region, $(1 + X)^{d_R} - 1$ counts a nonempty set of active slots at one right coordinate. Coefficient extraction conditions the regional slots to contain exactly the r supported left vertices. Matrix multiplication preserves the accumulator state across regions. The rank argument from Theorem 7.7 supplies the projective factor. $\square$

7.6 Finite-field EC

We now combine labeled two-sided regular incidence with the convolution in (9). The full-field feedback distribution preserves a reduced state with $m + 1$ classes. State $j < m$ records exactly j trailing zero outputs. State m means that the entire feedback state is zero.

The next transfer counts upper bounds rather than exact transition probabilities. Let z mark a nonzero output. Let v mark a zero event that imposes one equation in fresh randomness. For $b \in \{0, 1\}$, where b records whether the expander coordinate is occupied, define matrices $\mathbf{P}_b(z, v)$ by

$$(\mathbf{P}_b)_{j,0} = z, \quad (\mathbf{P}_b)_{j,j+1} = v \quad (0 \leq j < m), \quad (119)$$

$$(\mathbf{P}_0)_{m,m} = 1, \quad (\mathbf{P}_1)_{m,0} = z, \quad (\mathbf{P}_1)_{m,m} = v. \quad (120)$$

If the convolution state is nonzero, a zero output fixes one affine equation in the fresh vector α_t . Its probability is $1/p \leq 1/s$. If the state is zero and the coordinate is occupied, a zero expander

symbol fixes one equation in labels used only at that coordinate. Its probability is at most $1/s$. All other transition probabilities are bounded by one.

Define the regional matrix

$$\mathbf{R}_{\ell, d_R, r}^{\text{EC}}(z, v) := \binom{k}{r}^{-1} [X^r] \left(\mathbf{P}_0(z, v) + ((1 + X)^{d_R} - 1) \mathbf{P}_1(z, v) \right)^\ell. \quad (121)$$

Theorem 7.9 (Two-sided regular finite-field EC trace). *Independently sample a labeled two-sided regular expander $\mathbf{B}$ and the finite-field convolution $\mathbf{C}$ from (9). Then*

$$\Pr_{\mathbf{B}, \mathbf{C}} [\text{Bad}_L(\mathbf{BC})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L \sum_{a=0}^n [z^h v^a] \mathbf{e}_m^\top \mathbf{R}_{\ell, d_R, r}^{\text{EC}}(z, v)^{d_L} \mathbf{1} \min\{1, s^{r-1-a}\}. \quad (122)$$

Proof. Fix a support and its incidence pattern. Next fix one projective message line and a prescribed zero-nonzero output trace. Expose labels and feedback vectors in time order. Conditional on any realized past consistent with the prescribed trace prefix, each marked zero has probability at most $1/s$. The fresh vector or coordinate labels used for that step are independent of the past. Multiplying these conditional bounds gives at most s^{-a} for a trace with a marks. Its mark count depends on the incidence and prescribed trace, not on the message values.

Union-bound the s^{r-1} lines on the support and cap the result at one. The lines use the same sampled code; no independence between lines is needed. Only after this cap do we average over incidence patterns and sum over prescribed traces. The regional coefficient in (121) averages the two-sided regular incidence. Matrix multiplication retains the convolution state across regions. $\square$

Full-field sampling makes each inner product with a nonzero state uniform in $\mathbb{F}_p$. The same trace upper bound also holds if every feedback coefficient is sampled independently from $\mathbb{F}_p^*$. To see this, fix the past and a nonzero state coordinate. Condition on the other fresh coefficients and the current expander symbol. A zero output determines at most one value of the remaining coefficient, so its conditional probability is at most $1/s$. This proves the bound used above, although the exact inner-product law need not be uniform. Fixed all-one feedback does not satisfy this argument.

Dependence on the field. The trace bound uses the field only through its order p . The projective message count contributes $(p-1)^{r-1}$. Each independent nonzero-label equation contributes at most $(p-1)^{-1}$, and each full-field feedback equation contributes $p^{-1} \leq (p-1)^{-1}$. The remaining change is the p -ary GV cutoff. Consequently, the same argument applies in prime fields and extension fields. It does not apply to a binary matrix that is merely reinterpreted over an extension field, because that matrix lacks the sampled field labels and feedback coefficients.

7.7 Singleton refinement

The trace in Theorem 7.9 treats every occupied right coordinate alike. In fact, a coordinate incident to exactly one supported message symbol cannot vanish: both that symbol and its edge label are nonzero.

Let $\mathbf{P}_{\text{sg}}$ equal $\mathbf{P}_1$ except that $(\mathbf{P}_{\text{sg}})_{m,m} := 0$. Replace (121) by

$$\widehat{\mathbf{R}}_{\ell, d_R, r}^{\text{EC}}(z, v) := \binom{k}{r}^{-1} [X^r] \left(\mathbf{P}_0 + d_R X \mathbf{P}_{\text{sg}} + ((1 + X)^{d_R} - 1 - d_R X) \mathbf{P}_1 \right)^\ell. \quad (123)$$

Theorem 7.9 remains valid with this refined matrix.

For the lowest-equation traces, one more combinatorial bound removes a positive-marker loss. Let U_r be the number of occupied right coordinates when a uniform r -subset of the $d_R \ell$ regional slots is selected. Put

$$\beta_r := \Pr[U_r \leq \lfloor r/2 \rfloor], \quad (124)$$

$$J_r := 1 + \left\lfloor \frac{r-1}{m} \right\rfloor, \quad b_r := \left\lceil \frac{n-L-(r-1)}{\ell} \right\rceil - 2J_r. \quad (125)$$

Lemma 7.10 (Singleton-free regions). *For the ensemble of Theorem 7.9, let E_r be the event that some weight- r message has output weight at most L and a realized trace with at most $r-1$ fresh equations. If $1 \leq b_r \leq d_L$, then*

$$\Pr_{\mathbf{B}, \mathbf{C}}[E_r] \leq \binom{k}{r} \binom{d_L}{b_r} \beta_r^{b_r}. \quad (126)$$

If $b_r > d_L$, the event E_r is impossible.

Proof. Fix a support of size r and its incidence pattern. Consider any message on that support whose realized trace witnesses E_r . Every later entry into the all-zero convolution state requires m consecutive zero outputs from a nonzero state. Those outputs impose m fresh equations. A trace with at most $r-1$ equations therefore has at most J_r maximal all-zero-state intervals.

Such an interval can meet at most two regions containing a singleton coordinate. Otherwise it encounters a singleton inside a fully traversed region and must leave the all-zero state. If B regions are singleton-free, the all-zero-state intervals contain at most $(2J_r + B)\ell$ positions. All remaining zero outputs impose fresh equations. Thus

$$n - L \leq (2J_r + B)\ell + r - 1,$$

which implies $B \geq b_r$. Regional permutations are independent, and a singleton-free region has at most $\lfloor r/2 \rfloor$ occupied coordinates. The necessary incidence event does not depend on the message values or on which trace witnesses E_r . Union-bound this event over the required regions and the $\binom{k}{r}$ supports to obtain (126). $\square$

Apply this lemma before summing over traces with $a \leq r-1$. For traces with $a \geq r$, retain the refined transfer bound. The lemma bounds an existence event, not the sum of the individual low-equation trace bounds in (122).

This lemma is needed for the degree-22, memory-12 certificate. It removes slack in the structural branch; it does not assert that the ensemble fails when the bound is inconclusive.

7.8 From trace counts to finite bounds

The finite verifiers separate traces according to whether the number of fresh equations exceeds the projective message dimension. We give the two marker inequalities explicitly, because their marker domains differ. For one fixed support of size r , let

$$F_r(z, v) := \mathbf{e}_m^\top \widehat{\mathbf{R}}_{\ell, d_R, r}^{\text{EC}}(z, v)^{d_L} \mathbf{1}, \quad c_{r, h, a} := [z^h v^a] F_r(z, v).$$

The coefficients average trace counts over incidence patterns; they are not the probabilities of traces for one message. Define their two contributions by

$$S_r := \sum_{h=0}^L \sum_{a=0}^{r-1} c_{r, h, a}, \quad H_r := \sum_{h=0}^L \sum_{a=r}^n c_{r, h, a} s^{r-1-a}.$$

The refined trace theorem bounds the failure probability by $\sum_r \binom{k}{r} (S_r + H_r)$.

Lemma 7.11 (Markers for the two trace branches). *For $0 < z \leq 1$ and $0 < v \leq 1$,*

$$S_r \leq z^{-L} v^{1-r} F_r(z, v). \quad (127)$$

If additionally $v \geq 1/s$, then

$$H_r \leq s^{-1} z^{-L} v^{-r} F_r(z, v). \quad (128)$$

The two inequalities may use different marker pairs.

Proof. For $h \leq L$ and $a \leq r - 1$, one has $z^{h-L} v^{a-r+1} \geq 1$. For $h \leq L$ and $a \geq r$, the condition $sv \geq 1$ gives

$$s^{r-1-a} \leq s^{-1} v^{a-r} \leq s^{-1} z^{h-L} v^{a-r}.$$

Multiply by the nonnegative coefficient $c_{r,h,a}$ and sum. Extending either sum to all traces only increases its bound. $\square$

For small supports, the verifier evaluates F_r by exact regional coefficient extraction. For larger supports, it replaces that extraction by a positive marker $x > 0$. Put

$$\mathbf{M}(x, z, v) := \mathbf{P}_0 + d_R x \mathbf{P}_{\text{sg}} + ((1+x)^{d_R} - 1 - d_R x) \mathbf{P}_1.$$

Coefficient positivity and entrywise monotonicity of nonnegative matrix products give

$$F_r(z, v) \leq \binom{k}{r}^{-d_L} x^{-d_L r} \mathbf{e}_m^\top \mathbf{M}(x, z, v)^n \mathbf{1}. \quad (129)$$

After multiplying by $\binom{k}{r}$ and either branch factor, the only support-dependent factor is

$$g_r := \binom{k}{r}^{1-d_L} (x^{d_L} v)^{-r}.$$

The binomial coefficients have a concave logarithm, and $d_L \geq 1$. Thus g_r has a convex logarithm and is bounded on an integer interval $[u, w]$ by $g_u + g_w$. Multiplying that endpoint bound by $w - u + 1$ gives the block formula used by the verifier. At $r = k$, every right coordinate is occupied; evaluating $\mathbf{P}_1^n$ gives a direct upper bound without the coefficient marker.

These formulas also apply to the unrefined EC trace by replacing $\mathbf{P}_{\text{sg}}$ with $\mathbf{P}_1$, and to EA with its two-state input matrices. Where Lemma 7.10 is stronger, replace the low-equation existence bound before the trace sum, while keeping (128) for the high-equation branch.

8 Certified finite evaluation

The asymptotic theorems suppress polynomial factors and do not give a failure probability at one concrete length. Our finite certificates retain those factors and cover every nonzero message support. This section states the largest results and explains what the verifier checks. We call $-\log_2 U$ the *failure exponent in bits*: a larger value means a smaller upper bound U on bad sampling.

Table 6: Correspondence between verifier branches and mathematical bounds.

ensemble	branch	bound used by the verifier
Bernoulli EA	exact / block	(20) / (31)
Bernoulli EC	exact / block / dense	(40) / (42) / (43)
left-regular EA	exact / middle / dense	(58) / (59) / (60)
left-regular EC	exact / middle / dense	(63) / (64) / (65)
two-sided binary EC	exact / outer / central	(69) / (15) / Lemmas 6.11–6.12
labeled EA	exact / block	(118) / Lemma 7.11
labeled EC	exact / block / structural	(122) / Lemma 7.11 / Lemma 7.10

8.1 Verifier model

A certificate contains integer parameters, a partition of the support range, and decimal positive markers. A floating-point optimizer proposes the markers. The verifier parses each decimal as an exact input and recomputes the resulting expressions with Arb ball arithmetic through `python-flint`. A ball is a numerical interval together with a stated error radius; every computed ball encloses the exact real value.

The verifier checks four conditions:

1. the stated support intervals are ordered, disjoint, and cover the required range;
2. every marker belongs to the domain required by its inequality;
3. each matrix, coefficient bound, and Hamming-ball bound is evaluated with outward rounding;
4. the outward upper endpoint of the complete sum is below the claimed failure probability.

The optimizer is outside the trusted base. A poor marker only weakens a bound. Independent verification must also bypass stored numerical results. The degree-22 verifier supports a cache keyed by the certificate, verifier source, precision, and `python-flint` version. This key detects stale inputs; it does not prove that a cached numerical result is correct. The manifest therefore invokes that verifier with `--no-cache`.

Proposition 8.1 (Certificate soundness). *Suppose a verifier evaluates a proved first-moment upper bound, checks a partition of every support in $[1, k]$, and returns an Arb interval with upper endpoint U . Then the sampled ensemble satisfies $\Pr[\text{Bad}_L(\mathbf{G})] \leq U$.*

Proof. Each exact or relaxed support contribution is an upper bound by the theorem implemented for that interval. Outward rounding preserves every inequality. The checked partition makes the sum cover all nonzero message supports exactly once. The relevant first-moment theorem then gives the probability bound. $\square$

Table 6 gives the correspondence between verifier branches and proved bounds. In every row, block assembly uses Lemma 3.2. Exact branches evaluate the named nonnegative transfer directly. Binary marker branches use (15); the two field branches use Lemma 7.11 and (129).

The finite-field EC verifiers check the field order, regional dimensions, and complete support partition before numerical evaluation. This includes the separate sparse field branch in the degree-22 certificate. Each marker is checked against its branch’s domain before use. The two binary two-sided formats also have independent standard-library structural checkers. We do not claim that the latter checkers cover the other formats.

Table 7: Certified Bernoulli wrapped-EC profiles at rate one half.

C_0	memory m	expected degree $C_0 \ln n$	U	$-\log_2 U$
3.15	5	45.8517	$7.637727109 \cdot 10^{-7}$	20.320353
3.00	7	43.6683	$5.278295360 \cdot 10^{-7}$	20.853424
2.90	9	42.2127	$5.560776634 \cdot 10^{-7}$	20.778210
2.85	13	41.4849	$5.971032143 \cdot 10^{-7}$	20.675516

Table 8: Components of the intermediate binary two-sided EC certificates.

d_L/d_R	m	exact	outer	central	total	bits
14/7	6	$1.393702 \cdot 10^{-8}$	$3.872135 \cdot 10^{-33}$	$8.682238 \cdot 10^{-11}$	$1.402384 \cdot 10^{-8}$	26.0875
18/9	4	$2.637990 \cdot 10^{-7}$	$4.037754 \cdot 10^{-52}$	$3.493283 \cdot 10^{-8}$	$2.987318 \cdot 10^{-7}$	21.6746

8.2 Binary EC near the GV benchmark

The exact Bernoulli EC verifier gives several rate-one-half tradeoffs at $k = 2^{20}$, $n = 2k$, and $L = 230,741$. This cutoff is 0.9999819889 times the binary GV distance.

At rate 1/5, the profile with expected degree 77.3619, memory 9, and cutoff 1,274,032 reaches 99.99937% of the binary GV distance. Its failure probability is below $2^{-21.3200}$. These profiles show the finite degree–memory tradeoff before regular incidence is introduced.

8.3 Binary two-sided regular EC

The two-sided regular verifier proves (91). Exact regional transfers cover supports 1 through 16. The verifier uses 111 convex coefficient blocks outside the central range and 47 local-limit blocks inside it. A separate transfer checks the all-one message. At 192-bit precision, the outward-rounded contributions are at most

$$\begin{aligned}
U_{\text{exact}} &= 9.436470 \cdot 10^{-16}, \\
U_{\text{outer}} &= 1.633667 \cdot 10^{-10}, \\
U_{\text{central}} &= 1.382398 \cdot 10^{-12}, \\
U_{\text{full}} &< 10^{-341856}.
\end{aligned}$$

Their sum is below $1.6475000 \cdot 10^{-10}$, which gives 32.4990 failure bits. The outer range determines the final bound, not the central local-limit estimate.

The same verifier, with the odd-degree bound from Lemma 6.12, certifies the two intermediate profiles in Table 8. The column labeled “exact” sums the frozen exact-support transfers. The outer and central columns sum their respective support blocks.

The degree-14/7 verifier evaluates exact supports 1 through 24, 106 outer blocks, and 64 complemented central blocks. The degree-18/9 verifier evaluates exact supports 1 through 32, 102 outer blocks, and 76 complemented central blocks. In both profiles, support two is the largest exact term. The corresponding bounds are $1.108157 \cdot 10^{-8}$ and $2.583854 \cdot 10^{-7}$.

The degree-6/3, memory-79 verifier proves (94). Exact regional transfers cover supports 1 through 383. The verifier uses 103 convex coefficient blocks outside the central range and 28 complemented degree-three local-limit blocks inside it. The outward-rounded contributions are at

most

$$\begin{aligned} U_{\text{exact}} &= 2.829174 \cdot 10^{-7}, \\ U_{\text{outer}} &= 7.292690 \cdot 10^{-8}, \\ U_{\text{central}} &= 1.397617 \cdot 10^{-13}, \\ U_{\text{full}} &< 10^{-315674}. \end{aligned}$$

Their sum is below $3.558444 \cdot 10^{-7}$, which gives 21.4222 failure bits. The largest exact block covers supports 112 through 127 and is below $1.229988 \cdot 10^{-7}$.

The last exact block requires more dynamic range than one binary64 scale can provide. The verifier writes each nonnegative transfer as a sum of exponent layers. Entries in one layer share a dense mantissa matrix and a power-of-two scale. Matrix multiplication inflates each positive dot product upward. Separate layers retain small reachable entries without introducing an absolute error into structural zeros. Arb combines the scaled row sums with the message counts and output markers.

A checked-in JSON file for each profile fixes every positive marker and every block endpoint. A separate checker uses only integer arithmetic and decimal parsing from the Python standard library. It checks the parameter identities, marker domains, and a disjoint cover of supports 1 through k . The numerical verifier runs this checker before evaluating the stored markers. It does not invoke the floating-point optimizer during verification.

8.4 EA over a 127-bit prime field

The two-sided regular trace in Theorem 7.8 closes a large-field EA result at the exact floored GV cutoff.

Theorem 8.2 (Finite large-field EA certificate). *Let $p = 2^{127} - 1$, $d_L = 30$, $d_R = 15$, $\ell = 69,905$, $n = 2,097,150$, and $k = 1,048,575$. Let*

$$L = 1,032,064 = \lfloor n\delta_{p,\text{GV}}(1/2) \rfloor. \quad (130)$$

For the labeled two-sided regular EA ensemble,

$$\Pr_{\mathbf{B}}[\text{Bad}_L(\mathbf{BA})] < 9.499 \cdot 10^{-10} < 2^{-29.97}. \quad (131)$$

Proof. The verifier evaluates support sizes 1 through 64 with the exact regional trace. Twelve contiguous blocks cover $65 \leq r < k$; a proved convexity argument bounds every support inside a block from stored endpoint data. Full support is checked directly. At 192-bit precision, the three contributions are below $7.319 \cdot 10^{-30}$, $9.499 \cdot 10^{-10}$, and $1.036 \cdot 10^{-75}$. Proposition 8.1 completes the proof. $\square$

The degree is controlled by a dense-support term. This is why left regularity alone did not improve the large-field EA parameters in our scan. Two-sided regularity prevents right coordinates from being entirely empty at full message support.

8.5 EC over fields near 128 bits

The EC certificates combine the trace bound in Theorem 7.9, the singleton refinement, positive-marker coefficient bounds, and direct full-support evaluation. We first give a degree-memory frontier

over the 127-bit prime field $\mathbb{F}_{2^{127}-1}$. We then give two profiles over the 128-bit extension field $\mathbb{F}_{2^{128}}$. These are separate finite experiments: the fields, parameters, cutoffs, and sampled generators are fixed independently.

Theorem 8.3 (Finite $\mathbb{F}_{2^{127}-1}$ EC certificates). *Fix $p = 2^{127} - 1$ and rate $1/2$. For every row of Table 9 labeled EC, take d_L , d_R , n , L , and m from that row, and let $\ell = n/d_L$ and $k = n/2$. Independently sample the labeled two-sided regular expander and a finite-field convolution of memory m . Then*

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] < 2^{-b}, \quad (132)$$

where b is the displayed failure exponent.

Table 9: Certified recursive-map frontier over $\mathbb{F}_{2^{127}-1}$ at rate one half.

map	m	d_L/d_R	n	L	$d_L + m$	b
EA	1	30/15	2,097,150	1,032,064	31	29.9715
EC	3	28/14	2,097,144	1,032,062	31	27.6486
EC	4	26/13	2,097,134	1,032,057	30	71.4724
EC	6	24/12	2,097,144	1,032,062	30	39.5003
EC	12	22/11	2,097,150	1,032,064	34	27.2951

Proof. Each verifier separates the support range into exact regional bands, convex positive-marker blocks, and full support. The degree-22 verifier also uses Lemma 7.10 for structural traces through support 48. Its exact transfer covers supports 49 through 1400, and 26 convex blocks cover 1401 through $k - 1$. The complete outward-rounded bound is

$$6.071991718716554 \cdot 10^{-9} < 2^{-27.2951}.$$

The remaining three EC certificates use the same proved trace bounds without the additional low-support regional lemma where it is unnecessary. Their verified sums give the remaining rows. Proposition 8.1 converts each sum to the stated probability. $\square$

Theorem 8.4 (Finite $\mathbb{F}_{2^{128}}$ EC certificates). *Fix the field $\mathbb{F}_{2^{128}}$ and rate $1/2$. For each row of Table 10, independently sample the labeled two-sided regular expander and the finite-field convolution with the displayed parameters. Then*

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] < 2^{-b}, \quad (133)$$

where b is the displayed failure exponent. Each relative cutoff L/n is 98.415% of the rate-one-half $\mathbb{F}_{2^{128}}$ GV distance.

Table 10: Certified EC profiles over $\mathbb{F}_{2^{128}}$ at rate one half.

d_L/d_R	m	n	k	L	b
24/12	5	2,097,144	1,048,572	1,015,826	45.0317
26/13	4	2,097,134	1,048,567	1,015,822	135.2018

Proof. Both verifiers apply the singleton-refined trace bound with field order 2^{128} . For degrees 24/12 and memory 5, the outward-rounded exact-band, block, and full-support contributions sum to at most $2.780326 \cdot 10^{-14} < 2^{-45.0317}$. For degrees 26/13 and memory 4, they sum to less than $1.996089 \cdot 10^{-41} < 2^{-135.2018}$. The proof of Theorem 7.9 uses the field order through cardinalities and equation probabilities; it does not require that order to be prime. $\square$

The next theorem holds the degree and memory fixed while changing the field. It therefore isolates field-size scaling from the degree–memory tradeoff. Define

$$\eta := \frac{1,015,826}{2,097,144 \delta_{2^{128}, \text{GV}}(1/2)}.$$

Theorem 8.5 (Finite EC certificates near 128-bit fields). *For each row of Table 11, fix the field $\mathbb{F}_p$. Independently sample the labeled two-sided regular expander with degrees 24/12 and a finite-field convolution with memory 5. The code has length $n = 2,097,144$, dimension $k = 1,048,572$, and cutoff*

$$L = \lfloor \eta n \delta_{p, \text{GV}}(1/2) \rfloor.$$

Then $\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] < 2^{-b}$ for the displayed exponent b .

Table 11: Certified field-size profile for degrees 24/12 and memory 5.

field order p	cutoff L	failure exponent b
2^{127}	1,015,699	45.8357
2^{128}	1,015,826	45.0317
2^{129}	1,015,950	44.2357
2^{136}	1,016,774	38.7836

Proof. The verifier applies the singleton-refined trace bound from Theorem 7.9. It checks every nonzero message support with outward-rounded interval arithmetic. For field orders 2^{127} , 2^{128} , and 2^{129} , it uses one support partition and one set of positive markers. For field order 2^{136} , it locally retunes the structural output marker in five support bands. The verified failure exponents are, in table order,

$$45.8357737, \quad 45.0317394, \quad 44.2357386, \quad 38.7836609.$$

Proposition 8.1 gives the stated probability bounds. $\square$

The first row of Table 9 records EA as a memory-one recursive map because the accumulator retains y_{t-1} . It is not the memory-one instance of (9): EA fixes the feedback coefficient to one, whereas finite-field EC samples it from $\mathbb{F}_p$.

Among the EC rows, memory 4 and memory 6 minimize the simple work proxy $d_L + m$, both with value 30. Increasing memory to 12 reduces the expander degree to 22, but raises the proxy to 34. Thus a larger convolution helps the sparse regime, with diminishing returns. The diagnostic memory-two EC point also has left degree 30. It does not improve the expander degree over certified EA and requires additional random field arithmetic.

The proxy $d_L + m$ counts edges and retained field elements equally. It does not count the multiplications needed for random edge labels or feedback taps. Section 9 therefore treats it only as a guide for selecting profiles to benchmark.

The present analysis does not certify degree 20 with memory 21. At support 512, the exact structural upper bound is $2^{131.90}$. A positive upper bound is inconclusive; it is not evidence that the sampled ensemble has a low-weight codeword.

The diagnostic points $(m, d_L) = (1, 38)$ and $(2, 30)$ remain outside the certified frontier. Neither improves the implementation-facing choice: certified EA already uses degree 30 with a fixed memory-one recurrence, and the memory-two EC diagnostic has only a narrow floating-point margin. We do not use either point in a theorem or parameter recommendation.

9 Discussion and open questions

The exact enumerators change the interpretation of EA and EC parameters. There is no single mixing bottleneck. Dense messages already behave like the input to a random invertible map and attain the GV exponent. Sparse messages determine how many expander edges or convolution steps are needed before that regime becomes available.

What convolution buys. The wrapped convolution improves the sparse endpoint from $I_\delta = 1 - 2\sqrt{\delta(1-\delta)}$ for the accumulator to $I_{m,\delta}^w$ in (45). The latter tends to $1 - 2\delta$ as memory grows. It does not improve the dense exponent beyond the random-linear-code value. This explains the finite frontier: a few memory words can replace several expander edges, but large memory eventually has diminishing returns.

What regularity buys. Left regularity forces every message coordinate to contribute once per region. It reduces the variance in edge placement that dominates low-support terms. Two-sided regularity also eliminates empty right coordinates at full support. This property is essential to the large-field certificates. Over the binary field, odd right degree also prevents the all-one message from entering the expander kernel. Two-sided regularity substantially improves the binary EC profile, but it does not improve the weight-two EA bottleneck.

The certified frontier is not exhaustive. Remaining odd right degrees may yield additional tradeoffs. Measured encoding costs should eventually replace the proxy $d_L + m$. Near-regular right degrees are another possible way to avoid the binary parity obstruction, but they require a new point-mass bound.

Binary choices versus field values. The binary profiles target Silent OT, whose choices are bits. If the same matrix is viewed over $\mathbb{F}_{2^{128}}$, Lemma 6.14 still preserves its Hamming distance, but the map splits into 128 independent binary maps. Silent VOLE instead needs mixing between these components. Our field-valued profile supplies that mixing through ambient-field edge labels and full-field convolution coefficients.

Field size. Random nonzero edge labels are not cosmetic. They make the output law depend only on message support and enable the projective trace bounds. They also add one multiplication per edge. Finite-field EC uses full-field feedback for an exact uniform inner-product law. Independent nonzero feedback gives the same trace upper bound, as shown in Section 7. Other restricted tap distributions require a separate conditional point-mass bound; fixed all-one feedback is not covered by this argument.

The finite failure exponents can decrease with the field order when the target is a fixed fraction of the field-relative GV distance. Consider field order $p = 2^s$, with s bits per field element. At rate

one half,

$$\delta_{2^s, \text{GV}}(1/2) = \frac{1}{2} - \frac{1}{s} + O(s^{-2}).$$

Near $s = 128$, increasing s by one raises the GV-relative cutoff by about 126 output coordinates at our length. The low-equation structural branch receives no compensating field factor after its projective cap saturates. At a fixed absolute cutoff, increasing s does not weaken this branch; the field-suppressed branch only becomes smaller.

Table 12 illustrates the distinction for the degree-24/12, memory-5 certificate. It freezes the $\mathbb{F}_{2^{128}}$ support partition and positive markers. It then uses either the field-relative cutoff from Theorem 8.5 or the fixed $\mathbb{F}_{2^{128}}$ cutoff. The frozen exponent is the negative base-two logarithm of this floating-point bound. The certified column reports outward-rounded interval bounds where available. The smallest stored field marker is 2^{-127} , so every reused marker remains valid for the displayed orders.

Table 12: Field-size profile around 128-bit fields for degrees 24/12 and memory 5. A dash denotes a frozen-marker diagnostic without a separate interval certificate. The last column keeps the $\mathbb{F}_{2^{128}}$ cutoff fixed.

s	GV-relative cutoff	frozen bits	certified bits	fixed-cutoff bits
127	1,015,699	45.84	45.84	45.03
128	1,015,826	45.03	45.03	45.03
129	1,015,950	44.24	44.24	45.03
130	1,016,073	43.44	—	45.03
132	1,016,314	41.84	—	45.03
136	1,016,774	38.69	38.78	45.03
140	1,017,207	35.63	—	45.03
144	1,017,616	32.67	—	45.03

The certificates at $s = 127, 128, 129$ confirm a local loss of about 0.80 failure bits per additional field bit. At $s = 136$, local retuning improves the frozen exponent from 38.6903 to 38.7836 bits. The gain is only 0.0933 bits.

The structural branch controls this regime. At $s = 136$, the largest exact band covers message supports 129 through 160. Its frozen structural and field-suppressed exponents are -41.3863 and -162.9032 , respectively. Thus additional field suppression cannot compensate for the larger cutoff. At the fixed $\mathbb{F}_{2^{128}}$ cutoff $L = 1,015,826$, every displayed field order rounds to the same 45.03-bit diagnostic exponent. The decline comes from moving the GV-relative target, not from weaker mixing at the original distance.

This comparison does not include the certified $\mathbb{F}_{2^{127-1}}$ profile. That profile uses different degree-memory parameters and the full floored GV cutoff.

Finding a faster feedback distribution with an exact small state space remains open. A useful candidate must retain enough symmetry for a rigorous trace bound and use fewer field operations than full-field random taps.

Implementation boundary. The proved finite-field ensemble assigns an independent nonzero label to every expander edge. These labels act before the incident edges are summed. They therefore control cancellations at an expander coordinate.

Our reference implementation follows this structure. It materializes an independent balanced assignment for each region and gives every edge an independent nonzero label. It also samples every

convolution tap from the full field. A public seed expands all three sources of randomness.

The binary implementation materializes the two-sided regular expander and the wrapped convolution analyzed in Section 6. It is the Silent OT profile, not the Silent VOLE profile.

Our streaming implementation studies a different construction. Over odd-characteristic fields it uses a signed structured expander. Over characteristic two it uses compact full-index permutations. Both paths regenerate full-field convolution coefficients from a public seed and use field multiplications in the recurrence. These multiplications provide the component mixing needed by field-valued VOLE. The expander labels are not independent nonzero field elements, so we treat the streaming construction as heuristic rather than as a sample from the proved ensemble.

Our experiments also include unit edge coefficients as an ablation. They do not provide evidence of a defect at the tested sizes, but this observation is not a distance proof. Proving useful distance for structured or unit edge coefficients remains a separate problem.

EA-based pseudorandom correlation functions. The PCF construction in [BCG⁺22, Section 5.2] makes expander degree a per-output cost. In our notation, its noise vector has length n and one nonzero coordinate in each of t equal-sized blocks. Write $b := \lceil \log_2(n/t) \rceil$ for the bit length of a position within a block. A relaxed distributed comparison function lets the parties evaluate shares of this accumulated noise at a requested coordinate. An expander row of weight d requires d such evaluations. The leading work is about db pseudorandom-generator evaluations per party; the degree-two correlation construction has leading work quadratic in db [BCG⁺22, Section 5.4].

Our tighter EA bounds improve the code parameters underlying these costs. For example, at relative distance 0.05, the sufficient asymptotic Bernoulli degree threshold decreases from $4.938 \ln n$ to $1.773 \ln n$. This is a roughly 64% reduction in the degree coefficient, and hence in the corresponding coefficient of expected comparison-function work at fixed block size. It compares sufficient asymptotic distance conditions, not finite security margins or measured PCF runtimes. The original PCF estimates already use smaller, empirically motivated degrees; we do not claim the same percentage improvement over those estimates. Our finite bounds also supply explicit sampling guarantees: at $k = 2^{30}$ and $n = 5k$, expected degree 62 certifies relative distance 0.05 with failure below $2^{-20.23}$ (Table 3). This certificate uses Bernoulli rows, not rows conditioned to have weight 62.

Left-regular EA offers a second route: its independently sampled rows have fixed evaluation cost and retain the single accumulator. Our regional construction is the regular-row variant proposed in [BCG⁺22, Section 3.4]; Section 6 supplies finite distance bounds for it. These are bounds on expander regularity, distinct from the regularity of the PCF’s noise. The PCF security theorem still assumes EA-LPN with its specified noise distribution and a secure comparison primitive. Our contribution improves the distance guarantees supporting that assumption, not a reduction from minimum distance to full PCF security.

Cryptographic interpretation. A distance certificate bounds the probability that code generation produces a low-weight nonzero codeword. It does not by itself establish the security of a PCG, a PCF, or a silent protocol. Such an application must connect its attack model and noise distribution to the certified distance event. It must also account for any public structure not sampled by our ensemble.

For Silent VOLE over $\mathbb{F}_{2^{128}}$, the finite-field certificate supplies a distance premise for the independently labeled reference ensemble. A complete protocol theorem must still connect that premise to the chosen noise distribution and attack model. It must separately justify the structured streaming heuristic used by the implementation.

The following composition rule records exactly how the sampling term enters a protocol theorem.

Proposition 9.1 (Adding code-sampling failure). *Let $\mathbf{G}$ be a sampled generator, define the set $\mathcal{G}_{\text{good}} := \{G : d_{\min}(G) > L\}$, and suppose $\Pr[\mathbf{G} \notin \mathcal{G}_{\text{good}}] \leq U$. Let Fail be an event defined on the joint randomness of code generation and a protocol. If*

$$\Pr[\text{Fail} \mid \mathbf{G} = G] \leq \varepsilon \quad \text{for every } G \in \mathcal{G}_{\text{good}}, \quad (134)$$

then $\Pr[\text{Fail}] \leq \varepsilon + U$.

Proof. Condition on whether $\mathbf{G} \in \mathcal{G}_{\text{good}}$. The good branch contributes at most ε , and the bad branch contributes at most U . $\square$

The same additive term applies to a distinguishing advantage by replacing the bad-code branch with the trivial upper bound one. A Silent OT or Silent VOLE analysis must still prove the conditional bound ε for its noise distribution and attack model. This paper supplies only U .

Decoding and explicit constructions. The proofs are ensemble-existence and sampling statements. They do not give an efficient decoding algorithm. They also do not produce a deterministic family with the same distance. Both questions are orthogonal to the present first-moment method.

Certificate maintenance. The current artifacts use several specialized verifiers. A shared library for interval partitions, Arb matrices, and positive-marker bounds would make the trusted implementation smaller. An independent verifier in a second language would further reduce implementation risk.

The checked-in manifest maps each claimed numerical result to its verifier, inputs, SHA-256 digests, arithmetic precision, and expected failure exponent. The manifest driver checks these records and can execute all rigorous entries serially. The specialized verifiers remain part of the trusted base. A second implementation would reduce that trust further.

Conclusion. Exact input–output enumerators give substantially stronger guarantees for the original EA and EC ensembles. EA over every fixed finite field and fixed-memory wrapped binary EC approach GV with $O(n \log n)$ expected field operations and any fixed polynomial failure exponent. The degree constant controls that exponent. Expected cost $O(n \log^2 n)$ gives negligible failure at the same distance target. In the binary case, region-stratified regularity controls sparse messages and gives a concrete rate-one-half degree–memory frontier near the GV cutoff. In fields near 128 bits, random edge labels, projective grouping, and two-sided regularity control the additional union over message values. The finite-field analysis gives a separately certified field-mixing profile over $\mathbb{F}_{2^{127-1}}$. It also gives one fixed degree–memory profile over field orders 2^{127} through 2^{136} at a concrete cryptographic-scale length.

A Additional binary expander ensembles

The main text uses independent entries and region-stratified regular rows. This appendix records two intermediate choices. They are useful for comparing the effect of fixing a row or column weight while keeping the rest of the sampling rule unchanged.

A.1 Fixed column degree

In the fixed-column ensemble, each of the n columns is sampled independently and uniformly from the binary vectors of length k and weight d . Fix a message whose support has size r , and define

$$O_{r,d} := \sum_{b \geq 0} \binom{r}{2b+1} \binom{k-r}{d-2b-1}, \quad Z_{r,d} := \sum_{b \geq 0} \binom{r}{2b} \binom{k-r}{d-2b}. \quad (135)$$

The first quantity counts columns that meet the message support an odd number of times; the second counts those that meet it an even number of times. Thus $O_{r,d} + Z_{r,d} = \binom{k}{d}$.

Lemma A.1 (Fixed-column enumerator). *For $0 \leq u \leq n$, the expected number of weight- r messages whose expanded word has weight u is*

$$\binom{k}{r} \binom{n}{u} \left(\frac{O_{r,d}}{\binom{k}{d}} \right)^u \left(\frac{Z_{r,d}}{\binom{k}{d}} \right)^{n-u}. \quad (136)$$

Proof. For a fixed message, one output coordinate is nonzero exactly when the sampled column has odd intersection with the message support. This happens with probability $O_{r,d}/\binom{k}{d}$. The output coordinates are independent, so their weight is binomially distributed. The factor $\binom{k}{r}$ counts the messages of the stated weight. $\square$

The denominators in (136) are essential: without them the expression counts matrices instead of averaging over the ensemble.

A.2 Fixed row weight

In the fixed-row ensemble, the k rows are independent and uniform among the binary vectors of length n and weight d . Define

$$K_a^{(n)}(j) := \sum_{t \in \mathbb{Z}} (-1)^t \binom{j}{t} \binom{n-j}{a-t}. \quad (137)$$

This polynomial is the signed difference between the number of weight- a vectors that have even and odd intersection with a fixed j -coordinate set. It is often called a binary Krawtchouk polynomial.

Lemma A.2 (Fixed-row enumerator). *For $0 \leq u \leq n$, the expected number of weight- r messages whose expanded word has weight u is*

$$\binom{k}{r} \frac{\binom{n}{u}}{2^n \binom{n}{d}^r} \sum_{j=0}^n K_d^{(n)}(j)^r K_j^{(n)}(u). \quad (138)$$

Proof. For a fixed message, the expanded word is the binary sum of r independent uniform weight- d rows. The parity character indexed by a weight- j vector has expected value $K_d^{(n)}(j)/\binom{n}{d}$ on one row. Fourier inversion on the binary vector space gives the probability of each weight- u output. Multiplying by the $\binom{n}{u}$ outputs and $\binom{k}{r}$ messages gives (138). $\square$

The alternating sum in (138) is exact but poorly conditioned for numerical work. The following recurrence uses only nonnegative values. Let $\rho_{0,0} = 1$ and $\rho_{0,u} = 0$ for $u > 0$. If the current sum

Table 13: Certified fixed-row binary ensembles at relative distance 0.05.

scheme	s	row weight d	failure bound U	$-\log_2 U$
EA	20	31	$3.321605945 \cdot 10^{-7}$	21.521615
EA	25	35	$2.548539646 \cdot 10^{-7}$	21.903825
EA	30	39	$2.796467641 \cdot 10^{-7}$	21.769891
EC, $m = 21$	20	15	$5.160307473 \cdot 10^{-7}$	20.886039
EC, $m = 21$	25	17	$4.259825504 \cdot 10^{-7}$	21.162702
EC, $m = 21$	30	20	$4.830790936 \cdot 10^{-8}$	24.303165

has weight u , adding a uniform weight- d row gives weight $v = u + d - 2j$, where j is the size of the intersection. Therefore set

$$P_d(u, v) := \frac{\binom{u}{j} \binom{n-u}{d-j}}{\binom{n}{d}}, \quad j := \frac{u + d - v}{2}, \quad (139)$$

$$\rho_{r+1,v} := \sum_{u=0}^n \rho_{r,u} P_d(u, v). \quad (140)$$

The transition is zero unless j is an integer in the feasible intersection range. Then $\rho_{r,u}$ is the probability that a fixed weight- r message expands to weight u , and its expected shell count is $\binom{k}{r} \rho_{r,u}$.

For larger message weights, one may instead compare a weight- d row with an independent-entry row of density d/n . Conditioning the latter to have weight d costs

$$\vartheta_{n,d}^{-r}, \quad \vartheta_{n,d} := \binom{n}{d} (d/n)^d (1 - d/n)^{n-d}, \quad (141)$$

for r selected rows. Dense weights admit a separate Fourier bound. The needed fact is that a hypergeometric intersection count can be represented as a sum of independent Bernoulli variables [HP14]; Parseval's identity and Cauchy–Schwarz then compare the expanded word with a uniform word. This separation avoids paying the conditioning cost through the dense range.

A.3 Certified fixed-row examples

The positive recurrence, the conditioned comparison, and the dense Fourier bound give the following rate-1/5 certificates. Here $k = 2^s$, $n = 5k$, and $L = n/20$.

Theorem A.3 (Fixed-row binary certificates). *For each EA row of Table 13, independently sample every expander row as a uniform weight- d vector and apply the fixed accumulator. For each EC row, sample the same expander and independently sample a wrapped convolution of memory 21. The corresponding probability of Bad_L is at most the displayed value U .*

Proof. For EA, the verifier evaluates the nonnegative recurrence (140) at small supports. It applies the conditioning factor (141) at intermediate supports and the dense Fourier bound described above at large supports. For EC, the verifier composes the same regional shell calculation with the exact wrapped transfer (36). A conditioned Bernoulli comparison and an invertibility bound cover the remaining supports. In every row, the checked intervals partition $[1, k]$, and outward-rounded arithmetic gives the displayed upper endpoint. Lemma 3.2 and Proposition 8.1 complete the proof. $\square$

These values show that fixing the total row weight already removes much of the sparse-message loss. Region stratification improves further by forcing each row to spread across the entire word.

B The nonwrapping convolution

The main EC construction fixes the coefficient on the oldest of the m feedback symbols to one; the EC literature calls this choice *wrapping*. The earlier nonwrapping variant samples that coefficient with the other feedback coefficients. Both variants start from an all-zero feedback state. We give the nonwrapping enumerator to separate the two ensembles.

For an output prefix, let state $j \in \{0, \dots, m\}$ record the number of trailing zero output bits, capped at m . State m means that all m feedback bits are zero. From a state $j < m$, the feedback vector is nonzero, so a uniform binary tap vector makes the next output bit uniform. From state m , the feedback term is zero and the next output equals the next input.

Let $\mathbf{T}_m(X, Y)$ be the $(m+1) \times (m+1)$ matrix with nonzero entries

$$(\mathbf{T}_m)_{j,0} = \frac{1+X}{2}Y, \quad (\mathbf{T}_m)_{j,j+1} = \frac{1+X}{2} \quad (0 \leq j < m), \quad (142)$$

$$(\mathbf{T}_m)_{m,0} = XY, \quad (\mathbf{T}_m)_{m,m} = 1. \quad (143)$$

The exponent of X records input weight and the exponent of Y records output weight.

Lemma B.1 (Exact nonwrapping enumerator). *Let $\mathbf{e}_m$ select state m . The expected number of weight- u inputs that produce a weight- h output under a nonwrapping binary convolution is*

$$[X^u Y^h] \mathbf{e}_m^\top \mathbf{T}_m(X, Y)^n \mathbf{1}. \quad (144)$$

Proof. The two entries from an active feedback state average over the equally likely zero and one outputs. The two entries from the zero state copy the input. Matrix multiplication concatenates the transitions. The left vector selects the all-zero initial state, and the right vector sums over the final state. $\square$

For an independent Bernoulli input with nonzero probability q , averaging over the input marker gives a one-variable matrix $\mathbf{V}_{m,q}(Y)$:

$$(\mathbf{V}_{m,q})_{j,0} = Y/2, \quad (\mathbf{V}_{m,q})_{j,j+1} = 1/2 \quad (0 \leq j < m), \quad (145)$$

$$(\mathbf{V}_{m,q})_{m,0} = qY, \quad (\mathbf{V}_{m,q})_{m,m} = 1 - q. \quad (146)$$

Theorem B.2 (Nonwrapping EC first moment). *Sample $\mathbf{B} \leftarrow \mathcal{B}_{k,n,\rho}^{\text{Ber}}$ and an independent nonwrapping convolution $\mathbf{C}$ of memory m . With a_r as in (18),*

$$\Pr_{\mathbf{B}, \mathbf{C}}[\text{Bad}_L(\mathbf{BC})] \leq \sum_{r=1}^k \binom{k}{r} \sum_{h=0}^L [Y^h] \mathbf{e}_m^\top \mathbf{V}_{m,a_r}(Y)^n \mathbf{1}. \quad (147)$$

Proof. A fixed weight- r message has an independent Bernoulli expanded word with parameter a_r . Equations (145) and (146) give its exact output-weight generating function. Summing the low-weight coefficients and then the message shells gives an expected count of bad messages. Markov's inequality gives the claim. $\square$

For any $z \in (0, 1]$, the inner coefficient sum in (147) is at most

$$z^{-L} \mathbf{e}_m^\top \mathbf{V}_{m,a_r}(z) \mathbf{1}. \quad (148)$$

This is the same positive-marker relaxation used in the main proof. The difference between the wrapping and nonwrapping schemes is therefore confined to the transfer matrix and its boundary condition.

The finite certificates in this paper target the wrapped construction. The matrices above reduce a same-parameter comparison with nonwrapping EC to a finite numerical calculation, but we make no finite nonwrapping claim.

C Endpoint analysis for wrapped EC

This appendix proves the sparse and dense claims used in Lemma 5.3. It also explains why one set of constants controls a whole interval of input probabilities. Fix $m \geq 1$ and write $\varrho_m(a, z)$ for the largest absolute eigenvalue of $\mathbf{N}_{m,a}(z)$.

At $(a, z) = (0, 1)$, the reduced state chain has two closed classes. The active class consists of states $0, \dots, m-1$, and state m is the all-zero class. Define

$$c_m := \frac{1}{2^m - 1}, \quad \alpha_m := \frac{2^{m-1}}{2^m - 1}. \quad (149)$$

Within the active class, the stationary probability of state j is $\alpha_m 2^{-j}$. Thus c_m is the probability of the boundary state $m-1$, and α_m is the mean emitted weight in one active step.

Set $z = e^{-\theta a}$ for $\theta \geq 0$. Restricting the first-order perturbation of $\mathbf{N}_{m,a}(e^{-\theta a})$ to the two closed classes gives

$$\mathbf{G}_m(\theta) := \begin{pmatrix} -c_m - \theta \alpha_m & c_m \\ 1 & -1 \end{pmatrix}. \quad (150)$$

The first row represents the active class and the second the all-zero class. The off-diagonal entries are the first-order rates at which a small input probability moves between the classes. The marker contributes the additional rate $\theta \alpha_m$ while the chain is active.

The larger eigenvalue of (150) is

$$g_m(\theta) = -\frac{1 + c_m + \theta \alpha_m}{2} + \frac{\sqrt{(1 - c_m - \theta \alpha_m)^2 + 4c_m}}{2}. \quad (151)$$

The next lemma supplies the uniform remainder needed by the sparse-message argument. It is stated for a bounded marker parameter because the proof of Lemma 5.3 fixes one value of θ .

Lemma C.1 (Two-class perturbation). *Fix $m \geq 1$ and $0 \leq \Theta < \infty$. There are constants $a_0, K > 0$, depending only on m and Θ , such that*

$$\left| \ln \varrho_m(a, e^{-\theta a}) - a g_m(\theta) \right| \leq K a^2 \quad (152)$$

for $0 < a \leq a_0$ and $0 \leq \theta \leq \Theta$. The positive Perron eigenvector can be normalized so that the ratio of its largest coordinate to its smallest coordinate is at most K on the same domain.

Proof. Put $\mathbf{P}(a, \theta) := \mathbf{N}_{m,a}(e^{-\theta a})$. Its entries are analytic in (a, θ) near $\{0\} \times [0, \Theta]$. After decreasing a_0 if necessary, Taylor's theorem on the resulting compact set gives

$$\mathbf{P}(a, \theta) = \mathbf{P}_0 + a \mathbf{P}_1(\theta) + \mathbf{E}(a, \theta), \quad \|\mathbf{E}(a, \theta)\| \leq K_0 a^2, \quad (153)$$

for a constant K_0 and any fixed matrix norm.

The eigenvalue one of $\mathbf{P}_0$ is semisimple with multiplicity two. Its right one-eigenspace is spanned by the indicator vectors of the active and all-zero closed classes. All remaining eigenvalues have modulus less than one. Let $\mathbf{U}$ contain those two indicator vectors, and let $\mathbf{V}$ contain the dual stationary left eigenvectors, normalized by $\mathbf{V}^\top \mathbf{U} = \mathbf{I}_2$. Direct substitution of the transitions in (37)–(39) gives

$$\mathbf{V}^\top \mathbf{P}_1(\theta) \mathbf{U} = \mathbf{G}_m(\theta). \quad (154)$$

Use the spectral projection of $\mathbf{P}_0$ to decompose the state space into the two-dimensional one-eigenspace and its invariant complement. In this basis, the lower-right block of $\mathbf{P}_0 - \lambda \mathbf{I}$ is uniformly invertible whenever $|\lambda - 1|$ is sufficiently small. The Schur complement of that block, together with (153), is

$$(1 - \lambda) \mathbf{I}_2 + a \mathbf{G}_m(\theta) + \mathbf{R}(a, \theta, \lambda), \quad \|\mathbf{R}(a, \theta, \lambda)\| \leq K_1 a^2. \quad (155)$$

The constants are uniform for $\theta \in [0, \Theta]$. The eigenvalues of $\mathbf{G}_m(\theta)$ are separated by

$$\sqrt{(1 - c_m - \theta \alpha_m)^2 + 4c_m} \geq 2\sqrt{c_m} > 0.$$

Therefore the two eigenvalues near one are $1 + a\mu_i(\theta) + O(a^2)$, where the $\mu_i(\theta)$ are the eigenvalues of $\mathbf{G}_m(\theta)$. The remaining eigenvalues stay a fixed distance inside the unit circle.

Decrease a_0 so that $a_0 < 1$. For $0 < a \leq a_0$, the nonnegative matrix $\mathbf{P}(a, \theta)$ is irreducible. Its Perron eigenvalue is therefore the branch associated with the larger eigenvalue $g_m(\theta)$. The two eigenvalues of $\mathbf{G}_m(\theta)$ are distinct throughout the compact interval because $c_m > 0$. The Schur-complement equation also shows that the normalized Perron eigenvector converges uniformly to the lift of the positive eigenvector of $\mathbf{G}_m(\theta)$. Its coordinates are consequently bounded above and away from zero. Finally, $\ln(1 + ag_m(\theta) + O(a^2)) = ag_m(\theta) + O(a^2)$, with a uniform constant. This proves both claims. $\square$

Lemma C.1 gives

$$\ln \varrho_m(a, e^{-\theta a}) = ag_m(\theta) + O(a^2). \quad (156)$$

Substituting this marker into (44) gives

$$\liminf_{a \downarrow 0} \frac{J_{m,\delta}^w(a)}{a} \geq \sup_{\theta \geq 0} \{-\delta\theta - g_m(\theta)\}. \quad (157)$$

Put $x := \delta/\alpha_m$. Direct one-variable optimization yields

$$\sup_{\theta \geq 0} \{-\delta\theta - g_m(\theta)\} = (\sqrt{1-x} - \sqrt{c_m x})^2 \quad (158)$$

$$= \left(\sqrt{1-2\delta+2^{1-m}\delta} - \sqrt{2^{1-m}\delta} \right)^2 = I_{m,\delta}^w. \quad (159)$$

This proves the sparse exponent in (46).

We next justify the uniform probability bound that the asymptotic distance proof needs. For a nonnegative matrix and a positive right eigenvector v with eigenvalue λ , componentwise comparison gives

$$\mathbf{e}_m^\top \mathbf{N}_{m,a}(z)^n \mathbf{1} \leq \frac{\max_i v_i}{\min_i v_i} \lambda^n. \quad (160)$$

Fix $\varepsilon > 0$ and choose $\theta \geq 0$ attaining the supremum in (159). Then Lemma C.1 bounds the coordinate ratio of the Perron eigenvector for all sufficiently small a . Equations (156) and (160), together with the marker inequality, give constants $K, a_0 > 0$ such that

$$\Pr[H_{n,a} \leq \lfloor \delta n \rfloor] \leq K \exp\{-na(I_{m,\delta}^w - \varepsilon)\} \quad (161)$$

for all $n \geq 1$ and $a \in (0, a_0]$.

For input probabilities bounded away from zero, the matrix at $z = 1$ is a finite irreducible Markov transition matrix. Its stationary law satisfies

$$\pi_j = \pi_0 2^{-j} \quad (1 \leq j < m), \quad \pi_m = \pi_{m-1}, \quad \pi_0 = 1/2. \quad (162)$$

Consequently, the derivative of $\ln \varrho_m(a, z)$ with respect to $\ln z$ at $z = 1$ is $1/2$. Because $\delta < 1/2$, a marker slightly below one gives a positive lower-tail exponent. Continuity and a finite cover of every closed interval $[a_*, 1/2]$ give constants that are uniform throughout that interval. The rate function is also continuous there: the all-zero self-loop gives $\varrho_m(a, z) \geq 1 - a \geq 1/2$, so its defining objective is at most $\delta \ln z + \ln 2$. This tends uniformly to $-\infty$ as $z \downarrow 0$, allowing the optimization to be restricted to a compact interval of positive markers.

Finally, when $a = 1/2$, every row of $\mathbf{N}_{m,1/2}(z)$ sums to $(1 + z)/2$. Hence

$$\varrho_m(1/2, z) = \frac{1 + z}{2}. \quad (163)$$

Optimizing $\delta \ln z - \ln((1 + z)/2)$ gives $\ln 2 - h(\delta)$ and proves (47).

D Reproducing the finite certificates

The repository contains the certificate files, verifiers, manifest, and tests used for the finite claims. Run the following commands from the `expander_codes` directory. The commands are intentionally serial: each verifier can use substantial processor time and memory.

Install the tested dependency versions with

```
python -m pip install -r requirements.txt
```

The manifest records the tested Python and package versions. Its text-file digests normalize CRLF line endings to LF; all other bytes remain unchanged. To check every input digest, command, claim label, and dependency, run

```
python scripts/verify_manifest.py --strict-versions --list
```

To replay the principal results or every rigorous entry, respectively, run

```
python scripts/verify_manifest.py --run headline
```

```
python scripts/verify_manifest.py --run all
```

The driver launches one subprocess at a time and reports the elapsed time for each entry. The degree-6/3 exact transfer is tagged `slow`; its final band can take several minutes and use substantially more memory than the other binary checks.

Tests. The full test suite is

```
python -m unittest discover -s scripts -p "test_*.py"
```

Binary Bernoulli EA. The three rows of Table 3 are checked together by

```
python scripts/ea_certificate.py table \
  scripts/certificates/ea-k20-r5-d005-w50-s20.json \
  scripts/certificates/ea-k25-r5-d005-w56-s20.json \
  scripts/certificates/ea-k30-r5-d005-w62-s20.json
```

On PowerShell, replace each trailing backslash by a backtick.

Regular binary ensembles. Use the explicit rate-one-half parameters in Theorem 6.13:

```
python scripts/regular_ea_certificate.py --k 1048576 \
  --regions 64 --region-length 32768 --cutoff 230718
python scripts/regular_ec_certificate.py --k 1048572 \
  --regions 28 --region-length 74898 --cutoff 230730 --memory 9
```

Each two-sided binary EC claim uses a frozen marker file. Check its structure, then run its numerical verifier:

```
python scripts/check_binary_biregular_ec_certificate.py \
  results/binary_biregular_ec_rate_half_d10_m15_gv.json
python scripts/binary_biregular_ec_certificate.py verify \
  results/binary_biregular_ec_rate_half_d10_m15_gv.json
python scripts/check_binary_biregular_ec_certificate.py \
  results/binary_biregular_ec_rate_half_d14_m6_gv.json
python scripts/binary_biregular_ec_certificate.py verify \
  results/binary_biregular_ec_rate_half_d14_m6_gv.json
python scripts/check_binary_biregular_ec_certificate.py \
  results/binary_biregular_ec_rate_half_d18_m4_gv.json
python scripts/binary_biregular_ec_certificate.py verify \
  results/binary_biregular_ec_rate_half_d18_m4_gv.json
python scripts/check_binary_biregular_ec_certificate.py \
  results/binary_biregular_ec_rate_half_d6_m79_gv.json
python scripts/binary_biregular_ec_certificate.py verify \
  results/binary_biregular_ec_rate_half_d6_m79_gv.json
```

The structural checker uses only the Python standard library. It validates the parameter relations, marker domains, and support partition without loading the optimizer or the numerical verifier.

Finite-field EA. The certificate in Theorem 8.2 is checked by

```
python scripts/prime_field_biregular_ea_certificate.py verify \
  results/prime_field_biregular_ea_p127_rate_half_d30_gv.json
```

Finite-field EC. The four EC rows of Table 9 and the $\mathbb{F}_{2^{128}}$ results in Theorem 8.4 use the following verifier and certificate pairs:

```
python scripts/prime_field_biregular_ec_certificate.py verify \
  results/prime_field_biregular_ec_p127_rate_half_d28_m3_gv.json
python scripts/prime_field_biregular_ec_singleton_certificate.py verify \
  results/prime_field_biregular_ec_p127_rate_half_d26_m4_singleton_gv.json
python scripts/prime_field_biregular_ec_d24_m6_certificate.py verify \
  results/prime_field_biregular_ec_p127_rate_half_d24_m6_singleton_gv.json
python scripts/prime_field_biregular_ec_d22_m12_certificate.py verify --no-cache \
  results/prime_field_biregular_ec_p127_rate_half_d22_m12_region_gv.json
python scripts/prime_field_biregular_ec_singleton_certificate.py verify \
  results/finite_field_biregular_ec_gf128_rate_half_d26_m4.json
python scripts/prime_field_biregular_ec_singleton_certificate.py verify \
  results/finite_field_biregular_ec_gf128_rate_half_d24_m5_40bit.json
python scripts/prime_field_biregular_ec_singleton_certificate.py verify \
  results/finite_field_biregular_ec_gf127_rate_half_d24_m5_40bit.json
```

```
python scripts/prime_field_biregular_ec_singleton_certificate.py verify \
  results/finite_field_biregular_ec_gf129_rate_half_d24_m5_40bit.json
python scripts/prime_field_biregular_ec_singleton_certificate.py verify \
  results/finite_field_biregular_ec_gf136_rate_half_d24_m5_38bit.json
```

Each numerical verifier reads fixed decimal marker choices and recomputes the complete upper bound with outward rounding. Most bounds use Arb real balls. The degree-6/3 exact transfer uses upward-rounded exponent-layered binary64 matrices and passes their scaled row sums to Arb. Some older binary verifiers compute their markers at startup; the theorem-specific JSON certificates do not. The optimizer is not trusted: changing a marker can weaken a bound, but it cannot make the verifier accept an invalid inequality. The checked support partition is likewise part of verification.

The frozen-marker values in Table 12 are recorded in the following file:

```
results/finite_field_biregular_ec_gf128_rate_half_d24_m5_field_size_diagnostic.json.
```

The file identifies each interval certificate by its SHA-256 digest. It also marks the remaining calculations as floating-point diagnostics.

The machine-readable file `certificate_manifest.json` maps every finite theorem and table to its command, frozen inputs, arithmetic precision, and expected failure exponent. It hashes the complete trusted verifier source tree and every certificate used by a finite claim or diagnostic table. Entries marked `diagnostic` are not accepted as proofs. Frozen entries only invoke verification subcommands and never regenerate markers. A few legacy binary entries choose a marker at startup; their optimizer is untrusted because Arb checks the resulting inequality before the command succeeds.

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